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Effects of proprioceptive training in the recovery of patients submitted to meniscus surgery: systematic review and Meta-analysis

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Effects of proprioceptive training in the recovery of patients submitted to meniscus surgery: systematic review and Meta-analysis

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ABSTRACT

Objective: To evaluate the effects of proprioceptive training on rehabilitation of knee joint after meniscus surgery.

Methods: PubMed, EMBASE, The Cochrane Library, Web of science, China National Knowledge Infrastructure (CNKI), Technology Periodical Database (VIP), WanFang Data and China Biology Medicine (CBM) were searched until March 2021. Randomized controlled trials using proprioceptive training in patients who underwent meniscus surgery were included in this review. Study screening, data extraction and risk of bias evaluation were conducted independently by 2 researchers, and the Review Manager 5.3 software was used for meta-analysis.

Results: 9 articles with 453 patients met the inclusion criteria. Meta-analysis of outcomes of the proprioception test (MD= -1.73, 95% CI = [-2.98, -0.48], P<0.001), flexor muscle strength(SMD= 0.56,95%CI=[0.01, 1.11], P=0.05), and extensor muscle strength(SMD=0.31,95%CI=[0.06,0.56], P<0.05) revealed that proprioceptive training had significant treatment effects. However, there was no obvious advantage in improving the knee function score(SMD=0.85, 95%CI=[-0.13,1.84], P=0.09).

Conclusion: Proprioception training can effectively improve the proprioception, flexor muscle strength and extensor muscle strength of knee joint in patients with meniscus surgery. Proprioceptive training can be an effective supplement to conventional rehabilitation training. Limited by the quantity and quality of the included studies, the above conclusions need to be verified by more high-quality studies.

PROSPERO registration number CRD42020213201

Strengths and limitations of this study

► This study is the first systematic review and meta-analysis to evaluate the effects of proprioceptive training on rehabilitation of knee joint after meniscus surgery.

► The conclusion that proprioceptive training can be an effective supplement to conventional rehabilitation training could help patients after meniscus surgery to establish a better protocol for rehabilitation training.

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► The studies included in this review are generally of a low quality and this makes it difficult to draw firm conclusions.

INTRODUCTION

The meniscus is a fibrocartilaginous structure which is to bear weight, distribute pressure, lubricate and stabilize the joints and proprioception when they are in motion.[1] [2] Meniscal injuries are one of the most common diseases in orthopedics, with an annual average of 60-70 meniscal injuries per 100,000 cases concerning knee injuries.[2] This type of injury is generally classified into two etiologies.[3-5] One is caused by traumatic or sports injuries, when axial loads are transmitted directly to the flexed knee joint with rotation. The other is defined as a degenerative change of the cartilage, mostly found in middle-aged and elderly people. Arthroscopic partial meniscectomy has become a common treatment for meniscal injuries.[6 7] However, postoperative patients still suffer from a declining life quality, including pain, functional changes, proprioceptive deficits, neuromuscular and biomechanical changes, loss of quadriceps muscle strength.[8]

Proprioceptive training refers to a series of training methods, including practices of balancing, positioning, gait flexibility, agility and neuromuscular, which can improve patients' proprioception, knee function, and balancing impairments.[9-11] In recent years, proprioceptive training has been widely used in skeletal muscle rehabilitation which proved to be effective. The training process not only includes the rehabilitation after surgeries like anterior cruciate ligament reconstruction,[12 13] knee osteoarthritis,[14] hip and knee replacement,[10 15] but also contains the prevention of sports injuries.[16-18]

According to previous trials, no significant difference is shown between patients undergoing proprioceptive training and counterparts without such treatment after partial meniscectomy;[19 20] however, some studies have shown that proprioceptive training has good effects on knee rehabilitation.[21, 22] Considering the differences in the efficacy of proprioceptive training applied in the rehabilitation after meniscal injury, this study will not only meta-analyze the effect of proprioceptive training on the

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recovery of knee proprioception and muscle strength among postoperative meniscal-injured patients, but also provide reliable medical evidence for the efficacy of proprioceptive training on the recovery of knee function after meniscal injury.

METHODS

Protocol and registration

This systematic review and meta-analysis was conducted following the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA). The protocol of the current review was registered on the International Prospective Register of Systematic Reviews (PROSPERO, 2021, CRD42020213201).

Patient and public involvement

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Search strategy

A comprehensive search using the electronic databases of PubMed, EMBASE, Cochrane Library, Web of science, CNKI, VIP, WangFang Data and CBM was conducted in June 2021. Following keywords and their varies were used: proprioceptive training, proprioceptive neuromuscular facilitation, neuromuscular training, balance training, meniscectomy. The language was restricted in Chinese and English. Detailed search strategies based on guidance from the Cochrane handbook for the PubMed were showed in table 1 and would employ similar strategies for the other databases.

Study identification

All records were managed with Endnote X9. The republished articles and no English abstract articles were excluded. Two authors (XX and MJ) screened the studies and

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extracted the data independently according to the inclusion and exclusion criteria. The information of author, publication year, demographics of participants, intervention, the start time of intervention, training time and outcomes were recorded. Any disagreements were resolved by discussion or umpired with a third reviewer (WJ). After we reviewed relevant articles, eligibility criteria for this review based on PICOS frameworks (Population, Intervention, Comparison, Outcome and Study) were as follows: (1) patients with postoperative meniscal injury, whose race, nationality, and duration of disease are not limited; (2) Interventions: simple proprioceptive training and combined training with conventional rehabilitation were seemed as experimental groups; (3) conventional rehabilitation training were applied to control groups; (4) at least one of the outcomes of knee function or proprioception was reported; (5) randomized controlled trial (RCT) published in English or Chinese.

Risk of bias assessment

Two investigators independently assessed the quality of the literature using the PEDro scale.[23] 11 criteria were used in the PEDro scale, and each criterion was rated as "yes" or "no". Each yes would earn one score. However, the first question, “Whether we should make a detailed sampling criteria for the experiment” is excluded. The full mark is 10. A total PEDro score of ≥7 was considered as high quality, 5-6 as moderate quality, and ≤4 as low quality.[23] The scores were given independently by two researchers, and results would be rediscussed with a third researcher in case of disagreement.

Two researchers independently assessed the quality of included studies by using the tool for risk of bias—the Cochrane Collaboration Network.[24] We also assessed the following areas that may contain selection bias, allocation concealment, implementation bias, measurement bias, reporting bias, and other biases. Each area was classified into three categories: low risk, high risk and risk remained unclear.

Data extraction

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Two researchers independently selected the literature and extracted the data according to sampling criteria. The selection of the literature was based upon the relevance of the research topic. Irrelevant studies would be excluded by reading the title and abstract of the literature. Final decision would be made through a detailed reading of the full text. Data extraction was performed independently by two researchers, including: first author, year, age, sample size, type, time and frequency of intervention, and outcome indicators. Disagreements would be resolved through discussions.

Statistical analysis

Data included in the study were analyzed using Review Manager 5.3. Mean Difference (MD) was used for continuous variables. Standardized Mean Difference (SMD) was used for inconsistent measuring units to calculate the effect size and its 95% confidence interval (CI) for each combination.[25] The heterogeneity of treatment effects was examined by calculating the I^2 index. When I^2 was less than 50%, the combined effect was considered as mild heterogeneity and a fixed-effects model was utilized for meta-analysis; when I^2 was greater than 50%, the heterogeneity was considered to be high. Therefore, a random-effects model was used for meta-analysis and sensitivity analysis was performed to identify the source of heterogeneity. The level of Meta-analysis was set at α=0.05.

RESULTS

Study characteristics

As of March 2021, 142 relevant literatures were retrieved according to the sampling criteria, and 9 relevant articles were selected, 6 in Chinese[20-22 26-28] and 3 in English[19 29 30], with a total of 453 patients. A detailed selection process is shown in Figure 1.

All included literature entailed randomized controlled trials concerning the effect of proprioceptive training on knee function among patients with postoperative meniscal injuries. Four studies reported patients undergoing partial meniscectomy;[19 28-30] four

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studies reported patients undergoing meniscal repair;[21 22 26 27] one study did not touch upon the procedure;[20] seven studies reported the exclusion of patients with combined ACL injuries,[19 21 26-30] and 2 studies did not specifically report the procedure.[20 22] There are also various methods for assessing knee flexor and extensor muscle groups,[31 32] such as: isokinetic strength,[19 29] isometric strength,[20 26] peak torque strength,[30] and relative torque strength.[22] Specific literature characteristics are shown in table 2.

Quality assessment

The quality of the included literature was evaluated according to the PEDro quality assessment scale, and two obtained high quality,[19 30] six obtained moderate quality,[21 22 26-29] and one obtained low quality,[20] as detailed in table 3. The assessment concerning the risk of bias is shown in table 4.

Effects of proprioceptive training

Proprioceptive Function

Three studies reported the knee proprioceptive testing[20 21 26] in a total of 141 patients. All three studies assessed positioning sense using the same method, and MD was applied in the process of data merging. I^2 showed that $P<0.00001$, $I^2 = 92\%$ (Figure 2) with high heterogeneity. By using random effects model, we have found the data showed a statistical significance compared with control group (MD=-1.73,95% CI= [-2.98, -0.48], $P<0.001$), which could prove the positive effects brought by proprioceptive training for patients' knee proprioceptive sensation. After sensitivity analysis, one study[20] was identified as a source of high heterogeneity, which decreased significantly ($P<0.00001$, $I^2=18\%$) after being excluded. It was considered as a result of small sample size and improper methodology used in the research (Figure 3).

Flexor Muscle Strength

Five studies reported knee flexor muscle strength[19 22 26 29 30] in a total of 234 patients. The I^2 demonstrated P=0.002, $I^2=76\%$ in figure 4, with high heterogeneity. By adopting

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random effects model, we have found the data showed a statistical significance compared with control group (SMD= 0.56, 95% CI= [0.01,1.11], *P*=0.05), which could justify the positive effects on knee flexor (hamstring) muscle strength brought by proprioceptive training on patients. After sensitivity analysis, one study[26] was identified as the source of high heterogeneity, which decreased significantly (I^2= 0%, *P*=0.04) after being excluded. It was considered as a result of the seniority of the included patients and improper methodology in the research (Figure 5).

Extensor Muscle Strength

Six studies reported knee extensor muscle strength[19 20 22 26 29 30] in a total of 259 patients. I^2 showed *P*=0.22, I2=29% with low heterogeneity in figure 6. By using a fixed effects model, we have found the data showed a statistical significance compared with control group (SMD=0.31,95% CI= [0.06,0.56], *P*<0.05), which could justify that proprioceptive training could improve knee extensor (quadriceps) muscle strength in patients.

The Evaluation of Knee Functions

Five studies reported knee functions[19 21 27-29] in a total of 286 patients. The I^2 test showed P<0.00001, I^2=93% with high heterogeneity (Figure 7). By using random-effects model, we have found no significant difference between the two groups (SMD=0.85, 95% CI= [-0.13,1.84], *P*=0.09). After sensitivity analysis, 2 studies[19 29] were found to be the source of high heterogeneity, which decreased significantly (I^2= 0%, *P*<0.00001) after being excluded. It was considered partly because of the blank control as the control group (Figure 8).

Publication Biases

Considering the limited research for individual outcome indicators, no test for publication bias was performed in this study.

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DISCUSSION

This paper is the first systematic evaluation on the effects of proprioceptive training for the rehabilitation of patients undergoing meniscus surgery. It included 9 randomized controlled trials with 453 patients. It contained various levels of paper, including 2 high quality papers, 6 moderate quality papers and 1 low quality paper respectively. It is found the blinding and randomized sequential concealment were greatly biased while the incomplete data showed a minor bias. According to the results of meta-analysis, the included studies showed that the proprioceptive training could significantly improve knee proprioception, flexion and extension muscle strength in patients after meniscus surgery, while the positive effects to knee function was not significant. No adverse effects were reported in all studies.

Located in the outer third place of the meniscus, the proprioceptors could sense movements from the joints and their surrounding muscles, playing an important role in the maintenance of knee stability by regulating the amount of muscle strength around the knee joints through proprioceptive feedback.[9 20 21] Proprioception refers to the signal of body changes generated by proprioceptors in static or dynamic motion to specific parts of the body, which generally include the sense of static position, dynamic motion velocity, the direction of acceleration and the perception of joint pressure.[33] Proprioceptive training can help improve proprioception by restoring proprioceptive feedback mechanism of the knee joint,[17 34] which is consistent with our findings.

Partial meniscus impairment could also be found after arthroscopic meniscal surgery. Impaired proprioceptive feedback mechanism would predispose to reflex joint instability and irregular postural reflexes, leading to an increasing risk of degenerative disease or re-injury.[13 29 35] Proprioceptive feedback mechanism is a subjective perception of knee stability.[36] Once the knee proprioception is stimulated, signals are sent to the central nervous system, including the spinal cord, brainstem, and cerebral cortex, where they are analyzed and transmitted downward to protect the knee from injury by stimulating or inhibiting the corresponding muscles, adjusting posture and

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balancing in a timely manner.[13 37] On top of that, the protective system could also be activated by rapid protective reflexes via single or multiple synapses to limit or antagonize irregular knee movements and maintain knee stability.[38] The proprioceptive signals from the extremities, trunk, neck, combined with the sensory signals from the vestibular and visual systems will inform the central nervous system. All of the mentioned mechanisms work together to maintain knee stability and improve sportive performance.[13 30 39] In addition, it has been documented that conventional rehabilitation in patients after surgeries of meniscus injury can relieve pain and improve meniscus function,[40] which may justify the superiority of the conventional rehabilitation group over the proprioceptive training group in terms of the scores of knee function in one study.[29] It is evident that the conventional rehabilitation plays an important role in the recovery of patients after meniscal injury and constitutes an integral part of the postoperative rehabilitation.[29 41]

This Meta-analysis also has certain shortcomings. First, as seen in Figure 2, most of the included studies in this study had methodological flaws, mostly the lack of participant, therapist and assessor blinding. These factors may overestimate the efficacy of proprioceptive training.[42] Second, due to time inconsistencies of the included documents, this study generally focused on the effects of proprioceptive training after the surgery and thus lacked studies on midterm and long-term efficacy. Third, little literature could be found relating to this topic, to which different criteria were also used for analysis. Therefore, the conclusions were not supported by sufficient data. Fourth, the insufficiency of relevant studies on proprioceptive training and a lack of standardized scientific exercise prescription needed to be noted, such as the mode, time and intensity of sport.

CONCLUSIONS

This study showed that proprioceptive training, combined with conventional rehabilitation training, was significantly effective than conventional rehabilitation

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training in improving proprioception and the flexor-extensor muscle strength of the knee joint in patients undergoing surgeries designated for meniscal injury. However, in terms of scores, no significant difference had been found between the two groups in improving function of the knee joints. Given the limited number of included studies, a multicenter randomized controlled trial based on a large sample and a scientific methodology is still needed to further discuss the effect of proprioceptive training on functional recovery in patients after meniscal injury.

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Author Contributions

X.X.L., J.M., D.Z., J.W. and S.J. were responsible for conception and design and for the search procedure and check. X.X.L., J.M. and J.W. were responsible for extracting data and quality assessment. X.X.L. and J.M. made a great contribution in revising the . All authors made substantial contributions to conception and design and interpretation of the data, drafted the manuscript and gave final approval of the final version.

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Table 1 Search strategy for the PubMed database

Number	Search terms
#1	proprioceptive training [Title/Abstract]
#2	proprioceptive neuromuscular facilitation [Title/Abstract]
#3	neuromuscular training [Title/Abstract]
#4	balance training [Title/Abstract]
#5	#1 OR #2 OR #3 OR #4
#6	“meniscectomy” [Mesh]
#7	Meniscectomies [Title/Abstract]
#8	Arthroscopic Meniscectom* [Title/Abstract]
#9	Meniscectom*, Arthroscopic [Title/Abstract]
#10	Meniscal Resection*[Title/Abstract]
#11	Resection, Meniscal [Title/Abstract]
#12	#6 OR #7 OR #8 OR #9 OR #10 OR #11
#13	#5 AND #12

First author(Year)	Age（T/C）	Sample size（T/C）	Intervention（T/C）	Frequency	Duration	Outcomes

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Yuzhou Li 2014	32.0±6.25 26.5±6.75	12/13	CR+PT/CR	2 times per day, 5-7 times per week	8 weeks	topesthesia Isometric strength
Xiaoling Xiong 2018	29.2±5.12 28.3±5.36	15/15	CR+PT/CR	No description	8 weeks	Relative of torque
Weiyan Hung 2018	18-40	30/30	CR+BT/CR	No description	12 weeks	Lysholmscore topesthesia
Jianjiang Ouyang 2019	49.20±7.54 48.50±9.93	28/28	CR+PT/CR	1 times per day, 30 mins for each	12 weeks	Topesthesia Isometric strength
Wenjie Yu 2020	42.22±4.35 42.69±5.08	43/43	CR+PT/CR	2 times per day, 5-7 times per week	8 weeks	Lysholmscore rang of motion
Chendi Jiang 2021	40.85±5.47 40.53±5.46	24/24	CR+PT/CR	1 times per day, 6 times per week	8 weeks	Lysholmscore rang of motion
Xiaohui Zhang 2017	23.16±3.45 23.25±3.52	15/15	NT/CR	30 mins for each, 3 times per week	8 weeks	Lysholmscore Isokinetic strength
Ericsson 2008	45.4±3.2 45.9±3.2	28/28	NT/Blank control	5 times per week	8 weeks	peak torque
Hall 2015	42.8±5.4 43.2±5.6	31/31	NT/Blank control	30-45 mins for each, 1-3 times per week	12 weeks	KOOSscore Isokinetic strength

Table 2: Basic information of included documents. CR: conventional rehabilition; PT: proprioceptive training, NT: neuromuscular training, BT: balance training, KOOS: knee injury and osteoarthritis outcome score.

Study	1	2	3	4	5	6	7	8	9	10	11	Total score	Level
Yuzhou Li	×	×	×	×	×	×	×	√	√	√	√	4	Low
Xiaoling Xiong	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Weiyan Hung	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Jianjiang Ouyang	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Wenjie Yu	√	√	×	√	×	×	×	√	√	√	√	6	Fair

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Chendi Jiang	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Xiaohui Zhang	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Ericsson	√	√	√	√	×	×	√	√	√	√	√	8	High
Hall	√	√	√	√	×	×	√	√	√	√	√	8	High

Table 3: Evaluation of the quality of the included documents through PEDro. 1= inclusion exclusion criteria; 2 = randomized group; 3 = allocation concealment; 4 = similar at baseline; 5 = subject blinded; 6 = therapist blinded; 7 = assessor blinded; 8 = more than 85% of patient measures; 9 = intentional analysis; 10 = between-group analysis; 11 = presence of point measures for at least one outcome

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Study	Randomied method	Blinding	Allocation concealment	Data integrity	Selective repirting	Other biases
Yuzhou Li	Unknown	Unknown	Unknown	Complete	No	Unknown
Xiaoling Xiong	Unknown	Unknown	Unknown	Complete	No	Unknown
Weiyan Hung	Unknown	Unknown	Unknown	Complete	No	Unknown
Jianjiang Ouyang	Random number table	Unknown	Unknown	Complete	No	Unknown
Wenjie Yu	Random number table	Unknown	Unknown	Complete	No	Unknown
Chendi Jiang	Unknown	Unknown	Unknown	Complete	No	Unknown
Xiaohui Zhang	Random number table	Unknown	Unknown	Complete	No	Unknown
Ericsson	Random number table	Single blind	Sealed envelop	Complete	No	Unknown
Hall	Random number table	Single blind	Unknown	Complete	No	Unknown

Table 4: Evaluation of the risk of bias of the included documents.

Figure 1: Flow chart of literature selection

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Figure 2: Meta-analysis of proprioceptive tests

Figure 3: Meta-analysis of proprioceptive tests (after reducing heterogeneity)

Figure 4: Meta-analysis of the knee flexor muscle strength

Figure 5: Meta-analysis of the knee flexor muscle strength (after reducing heterogeneity)

Figure 6: Meta-analysis of the knee extensor muscle strength

Figure 7: Meta-analysis of knee function scores

Figure 8: Meta-analysis of knee function scores(after reducing heterogeneity)

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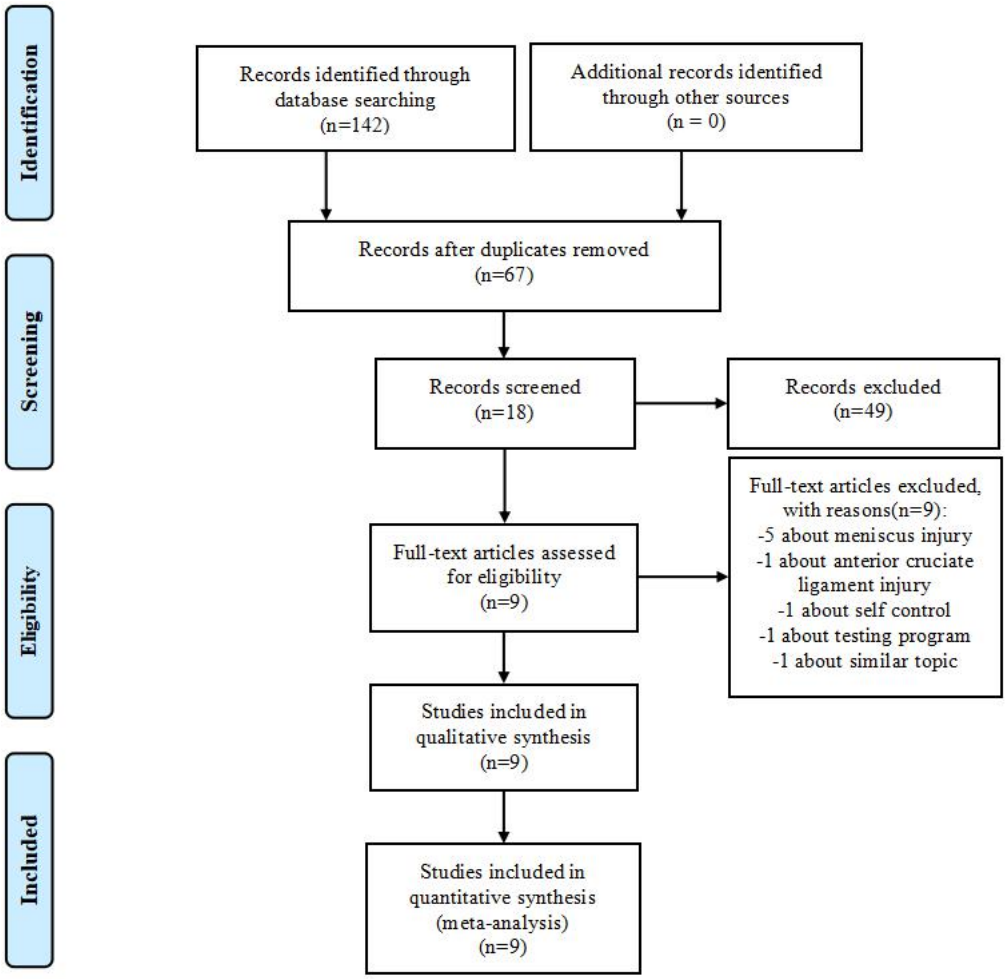


Figure 1: Flow chart of literature selection

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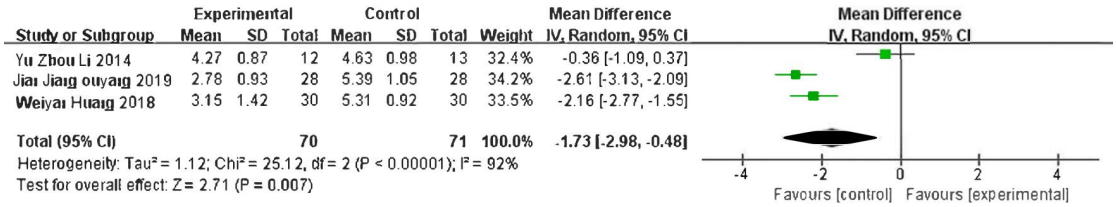


Figure 2: Meta-analysis of proprioceptive tests

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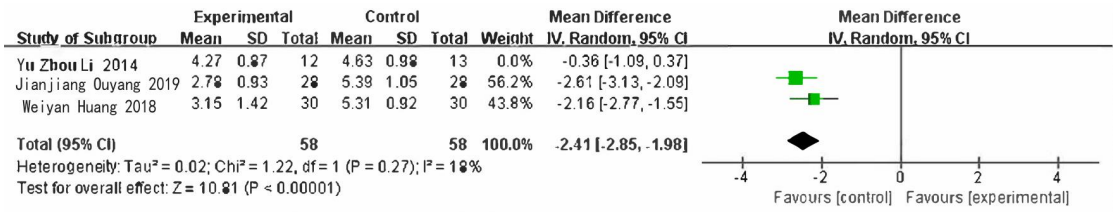


Figure 3: Meta-analysis of proprioceptive tests (after reducing heterogeneity)

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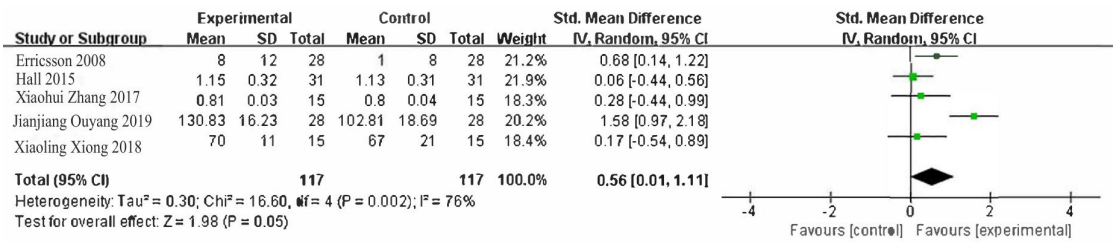


Figure 4: Meta-analysis of the knee flexor muscle strength

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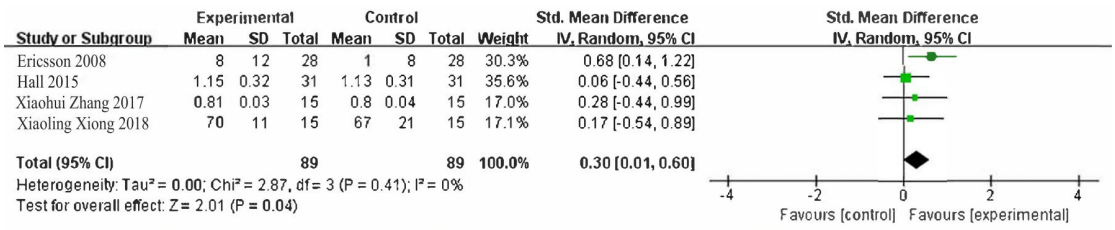


Figure 5: Meta-analysis of the knee flexor muscle strength(after reducing heterogeneity)

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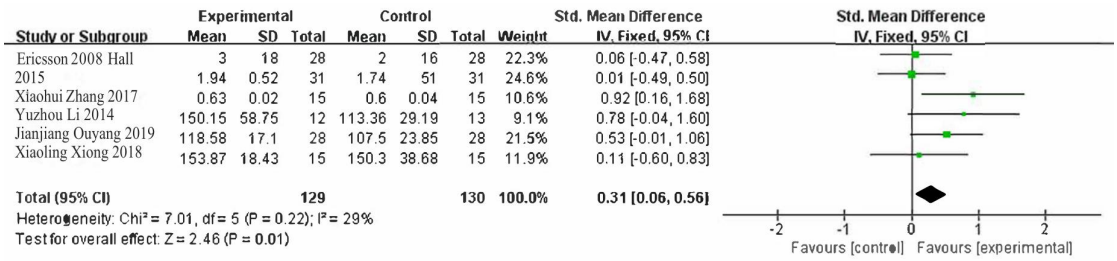


Figure 6: Meta-analysis of the knee extensor muscle strength

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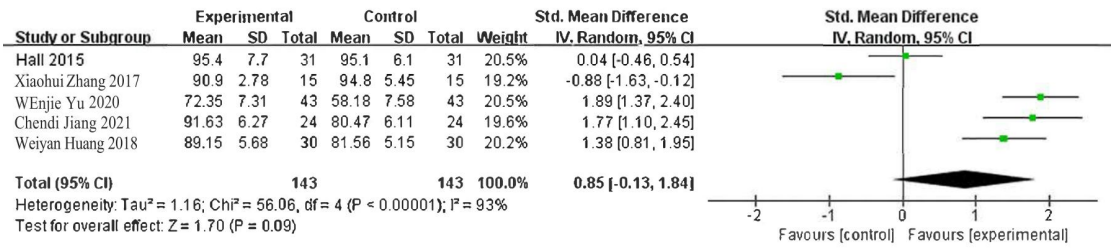


Figure 7: Meta-analysis of knee function scores

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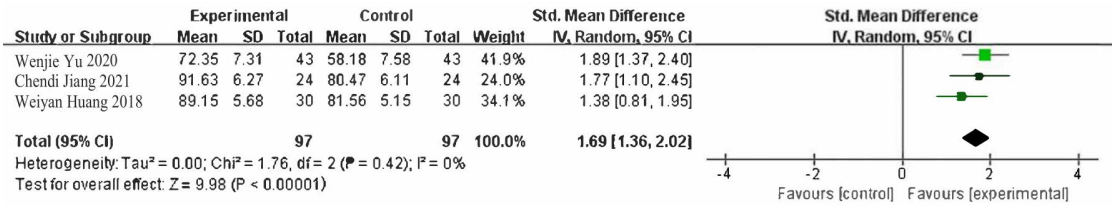


Figure 8: Meta-analysis of knee function scores(after reducing heterogeneity)

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			1
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			2
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			3-4
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			4-6
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			6-8
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7-8
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8
DISCUSSION			9-10
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	10
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			11
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	11

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

Effects of proprioceptive training in the recovery of patients submitted to meniscus surgery: systematic review and Meta-analysis

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Secondary Subject Heading:	Rehabilitation medicine, Sports and exercise medicine
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Effects of proprioceptive training in the recovery of patients submitted to meniscus surgery: systematic review and Meta-analysis

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ABSTRACT

Objective To evaluate the effects of proprioceptive training on rehabilitation of knee after arthroscopic partial meniscectomy (APM).

Design PubMed, EMBASE, The Cochrane Library, Web of science, China National Knowledge Infrastructure (CNKI), Technology Periodical Database (VIP), WanFang Data and China Biology Medicine (CBM) were searched until December 2021 for randomized controlled trials.

Participants Patients who have undergone APM for meniscus injury caused by traumatic tear.

Results A total of 9 studies were included in this study for meta-analysis, including 453 patients. Based on low quality evidence, meta-analysis of outcomes of the proprioception test (MD= -1.73, 95% CI= [-2.98, -0.48], $P< 0.001$) and extensor muscle strength of knee (SMD= 0.31, 95% CI= [0.06, 0.56], $P= 0.01$) revealed that proprioceptive training had significant treatment effects in patients with APM compared to conventional training. Proprioceptive training had treatment effects but no significant difference in improving the flexor muscle strength of knee (SMD= 0.56, 95% CI= [0.01, 1.11], $P= 0.05$) and function score of knee (SMD= 0.85, 95% CI= [-0.13, 1.84], $P= 0.09$) based on low quality evidence.

Conclusion: Adding proprioceptive training to conventional rehabilitation programs might be beneficial to promote function recovery for patients with APM based on low quality evidence. It is necessary to carry out more samples and higher quality large-scale studies to provide high evidence in the future.

PROSPERO registration number CRD42020213201

Strengths and limitations of this study

- ► This meta-analysis was preregistered and conducted in line with the Preferred Reporting Items for Systematic Reviews and Meta-analyses guidelines.
- ► This study is the first systematic review and meta-analysis to evaluate the effects of proprioceptive training on rehabilitation of knee after arthroscopic partial

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meniscectomy.

- ► Eight electronic databases were searched to provide a comprehensive range of studies.
- ► Differences in the control interventions, the time-course of treatment, the start time of intervention and assessment may increase heterogeneity and bias.

INTRODUCTION

Meniscal injuries are one of the most common diseases in orthopedics, with an annual average of 60- 70 meniscal injuries per 100, 000 cases concerning knee injuries.[1 2] Meniscus tears were detected in up to 80% of cases when people have magnetic resonance imaging scans to check the knee.[3] This type of injury is generally classified into two etiologies.[4-6] One is caused by traumatic or sports injuries, when axial loads are transmitted directly to the flexed knee joint with rotation. The other is defined as a degenerative change and often accompanied by degenerative changes in cartilage, mostly found in middle-aged and elderly people, which often involved in knee osteoarthritis.[1 3] Arthroscopic partial meniscectomy (APM) is among the most common procedures performed for their treatment.[1 7 8] Study shows that about 636 000 knee arthroscopy procedures are performed every year in the United States.[3] In addition, recent studies have shown that surgical treatment has been transferred toward facilitating meniscal repair to maintain meniscal tissue integrity whenever possible, prevent secondary cartilage degeneration and improve knee joint function.[1 9-11] However, postoperative patients experience pain and swelling leading to loss of range of movement, proprioceptive deficits, neuromuscular and biomechanical changes, decreased quadriceps femoris muscle strength, and further leading to dysfunction. The dysfunction especially in proprioception which perceives changes in the state of motion and initiates protection and muscular reflexes to keeps dynamic stabilizing are prone to increase the risk of degenerative disease or reinjury in long-term after surgery.[1 12-14]

Proprioception plays an important role in postoperative rehabilitation. The inputs from the sensory organs are processed in the brain and integrated with visual and

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vestibular information to generate a sense of position and movement through space. Proprioceptive training can stimulate the sensory organ to send specific signals to control the relevant muscles to maintain stability through specific dynamic movement exercises such as practices of balancing, positioning, gait flexibility, agility and neuromuscular.[15-17] In recent years, adding proprioceptive training to conventional rehabilitation has been widely used in postoperative rehabilitation of joint at different times, some on the first day after surgery, and some three months after surgery, which proved to be effective in improving proprioception and activity function.[1] Besides, benefits could be obtained after proprioceptive training for not only patients with anterior cruciate ligament reconstruction,[18 19] knee osteoarthritis,[20] hip and knee replacement,[16 21] but also athletes with the prevention of sports injuries.[22-24]

Proprioception training is beneficial to patients with APM. According to previous trials, some studies have shown that proprioceptive training has a significant improvement on knee rehabilitation for patients with APM.[25 26] However, no significant difference is shown between patients undergoing proprioceptive training and counterparts without such treatment after surgery.[27 28] Considering the differences in the efficacy of proprioceptive training applied in the rehabilitation, this study will not only meta-analyze the effect of proprioceptive training on the recovery of knee proprioception and muscle strength among patients with APM caused by traumatic tear, but also provide reliable medical evidence for the efficacy of proprioceptive training on the recovery of proprioception and function after APM.

METHODS

Protocol and registration

This systematic review and meta-analysis was conducted following the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-analyses(PRISMA). The protocol of the current review was registered on the International Prospective Register of Systematic Reviews(PROSPERO, 2021, CRD42020213201).

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Patient and public involvement

Patients and/ or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Search strategy

A comprehensive search using the electronic databases of PubMed, EMBASE, Cochrane Library, Web of science, CNKI, VIP, WangFang Data and CBM was conducted in December 2021. Following keywords and their varies were used: proprioceptive training, sensorimotor training, proprioceptive neuromuscular facilitation, neuromuscular training, balance training, meniscectomy. The language was restricted in Chinese and English. Detailed search strategies based on guidance from the Cochrane handbook for all above databases were showed in the supplemental appendix S1.

Study selection

All records were managed with Endnote X9. The republished articles and no English abstract articles were excluded. Two authors (XX and MJ) screened the studies and extracted the data independently according to the inclusion and exclusion criteria. The information of author, publication year, demographics of participants, intervention, the start time of intervention, training time and outcomes were recorded. Any disagreements were resolved by discussion or umpired with a third reviewer (WJ). After we reviewed relevant articles, eligibility criteria for this review based on PICOS frameworks (Population, Intervention, Comparison, Outcome and Study) were as follows: (1) Population: patients who have undergone APM for meniscus injury caused by traumatic tear, whose race, nationality, and duration of disease are not limited, patients who have undergone meniscal surgery combined with other procedures such anterior cruciate ligament reconstruction would be excluded; (2) Interventions: proprioceptive training alone or combined training with conventional rehabilitation were seemed as experimental groups; (3) Comparison: conventional

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rehabilitation training (including ankle pump movement, continuous passive motion, weight bearing exercise, strength training, etc.) were applied to control groups, control groups intervention did not include proprioceptive training; (4) Outcome: at least one of the outcomes of extensor, flexor muscle strength, knee function score or proprioception test was reported; (5) Study: randomized controlled trial (RCT) published in English or Chinese.

Risk of bias assessment

Two investigators independently assessed the quality of the literature using the PEDro scale.[29] Eleven criteria were used in the PEDro scale, and each criterion was rated as "yes" or "no". Each yes would earn one score. However, the first question, “Whether we should make a detailed sampling criteria for the experiment” is excluded. The full mark is 10. A total PEDro score of ≥ 7 was considered as high quality, 5- 6 as moderate quality, and≤ 4 as low quality.[29] The scores were given independently by two researchers, and results would be rediscussed with a third researcher in case of disagreement.

Data extraction

Two researchers independently selected the literature and extracted the data according to sampling criteria. The selection of the literature was based upon the relevance of the research topic. Irrelevant studies would be excluded by reading the title and abstract of the literature. Final decision would be made through a detailed reading of the full text. Data extraction was performed independently by two researchers, including: first author, year, age, sample size, type, time and frequency of intervention, and outcome indicators. Disagreements would be resolved through discussions.

Statistical analysis

Data included in the study were analyzed using Review Manager 5.3. Mean Difference (MD) was used for consistent measuring units. Standardized Mean Difference (SMD)

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was used for inconsistent measuring units to calculate the effect size and its 95% confidence interval (CI) for each combination.[30] The heterogeneity of treatment effects was examined by calculating the I^2 index. Interpretation of I^2 is as follows: 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity. When I^2 was less than 50%, the combined effect was considered as mild heterogeneity and a fixed-effect model was utilized for meta-analysis; when I^2 was greater than 50%, the heterogeneity was considered to be high. Therefore, a random-effect model was used for meta-analysis and sensitivity analysis was performed to identify the source of heterogeneity. Absence of blinding and differences in the control interventions, the time-course of treatment, the start time of intervention and assessment may increase heterogeneity. *P* value level of 0.05 was set for between-group differences.

GRADE Assessment

The GRADE Approach (Grading of Recommendations, Assessment, Development and Evaluations) was used to appraise and summarise the body of evidence.[31] GRADE is an internationally approved standard for managing complex evidence reviews. For results based only on RCTs certainty was initially considered as high. Thereafter, certainty could be rated down based on factors such as risk of bias, imprecision, inconsistency, indirectness and potential publication bias.[31 32]

RESULTS

Study characteristics

As of December 2021, 155 potentially relevant literatures were retrieved according to the sampling criteria, and 9 relevant articles were selected, 6 in Chinese[26-28 33-35] and 3 in English[27 36 37], with a total of 453 patients. A detailed selection process is shown in Figure 1.

All included literature entailed RCTs concerning the effect of proprioceptive training on knee function among patients with APM.[25-28 33-37] Seven studies reported the

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exclusion of patients with combined ACL injuries,[25 27 33-37] There are also various methods for assessing flexor and extensor muscle strength of knee,[38 39] such as: isokinetic strength,[27 36] isometric strength,[28 33] peak torque strength,[37] and relative torque strength.[26] There are also various methods for assessing function scores of knee such as: Lysholm scores[25 34-36] and Knee Injury and Osteoarthritis Outcome Score.[27] Specific study characteristics are shown in Table 1.

Quality assessment

The quality of the included studies was evaluated according to the PEDro quality assessment scale, most of all had methodological flaws in subject, therapist and assessor blinding. Two obtained high quality,[27 37] six obtained moderate quality,[25 26 33-36] and one obtained low quality,[28] as detailed in Table 2.

Effects of proprioceptive training

Proprioceptive test

Three studies with 141 patients reported the knee proprioceptive test.[25 28 33] All three studies assessed positioning sense using the threshold to detection of passive motion, and MD was applied in the process of data merging. I^2= 92% (Figure 2-a) with high heterogeneity. By using random effects model, pooling of the data showed a statistical significance compared with control group (MD= -1.73, 95% CI= [-2.98, -0.48], P< 0.001), which could prove the positive effects brought by proprioceptive training for patients' knee proprioception. After sensitivity analysis, the study[28] was identified as a source of high heterogeneity, which decreased significantly (P< 0.00001, I^2= 18%, Figure 2-b) after being excluded. It was considered as a result of small sample size and improper methodology used in the study.

Flexor muscle strength

Five studies with 234 patients reported the flexor muscle strength of knee.[26 27 33 36 37] The I^2= 76% in Figure 3-a, with high heterogeneity. By adopting random effects model,

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pooling of the data showed no statistical significance compared with control group (SMD= 0.56, 95% CI= [0.01, 1.11], *P*= 0.05). After sensitivity analysis, the study[33] was identified as the source of high heterogeneity, which decreased significantly (I^2= 0%, *P*= 0.04, Figure 3-b) after being excluded, which could justify the positive effects on knee flexor (hamstring) muscle strength brought by proprioceptive training. It was considered as a result of the seniority of the included patients and improper methodology in the study.

Extensor muscle strength

Six studies with 259 patients reported the extensor muscle strength of knee.[26-28 33 36 37] I^2 =29% with low heterogeneity in Figure 4. By using a fixed effects model, pooling of the data showed a statistical significance compared with control group (SMD= 0.31, 95% CI= [0.06, 0.56], *P*= 0.01), which could justify that proprioceptive training could improve knee extensor (quadriceps) muscle strength.

Function scores

Five studies with 286 patients reported the function scores of knee assessed by Lysholm score[27] and knee injury and osteoarthritis outcome score[25 34-36]. The I^2= 93% with high heterogeneity (Figure 5-a). By using random-effects model, pooling of the data showed no significant difference between the two groups (SMD= 0.85, 95% CI= [-0.13, 1.84], *P*= 0.09). After sensitivity analysis, two studies[27 36] were found to be the source of high heterogeneity, which decreased significantly (I^2= 0%, *P*< 0.00001, Figure 5-b) after being excluded one by one. It was considered partly because of the blank control as the control group.

GRADE approach level of evidence

Pooled results of proprioceptive test, flexor muscle strength of knee, extensor muscle strength of knee, function scores of knee comparing proprioceptive training group to conventional training group were considered of very low quality. See details of Grade

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Approach and conclusions in Table 3.

DISCUSSION

This study is the first systematic review on the effects of proprioceptive training for the rehabilitation of patients undergoing APM. Nine RCTs with 453 patients containing 2 high-quality studies, 6 moderate-quality studies and 1 low-quality study respectively were included. It was found that the blinding and randomized sequential concealment was greatly biased while the incomplete data showed a minor bias. According to the results of meta-analysis, the included studies showed that the proprioceptive training could significantly improve knee proprioception, flexion and extension muscle strength in patients after APM, while the positive effects to knee function scores was not significant. GRADE approach showed that all the pooled results comparing proprioceptive training group to conventional training group were considered of very low quality.

Proprioceptive training could significantly improve knee proprioception in patients after APM. Previous study[40] showed that patients with isolated meniscal tears had a significant proprioceptive deficit when compared to their uninjured contra-lateral knee and to the healthy subjects. Located in the outer third place of the meniscus, the proprioceptors could sense the condition of movement from the knee joints and their surrounding muscles, playing an important role in the maintenance of knee stability by regulating the amount of muscle strength around the knee joints through proprioceptive feedback.[15 25 28] Proprioception refers to the signal of body changes generated by proprioceptors in static or dynamic motion to specific parts of the body, which generally include the sense of static position, dynamic motion velocity, the direction of acceleration and the perception of joint pressure.[41] Proprioceptive feedback mechanism is a subjective perception of knee stability.[42] Once the knee proprioception is stimulated, signals are sent to the central nervous system, including the spinal cord, brainstem, and cerebral cortex, where they are analyzed and transmitted downward to protect the knee from injury by stimulating or inhibiting the corresponding muscles, adjusting posture

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and balancing in a timely manner.[19 43] After the meniscus injury partially or completely, the proprioception and neuromuscular control ability of knee are significantly decreased.[28 43] Partial meniscus impairment could also be found after APM. APM could not resulted in a significant improvement in proprioceptive function.[44] Impaired proprioceptive feedback mechanism would predispose to reflex joint instability and irregular postural reflexes, leading to an increasing risk of degenerative disease or re-injury.[19 36 45] Postoperative proprioceptive training could effectively restore the proprioceptive function, which is consistent with our results.

The results in this study show significant improvement for proprioceptive training group in extensor muscle strength of knee, but no differences in flexor muscle strength and function score of knee compared to conventional training group. Although the *P* value in improving flexor muscle strength is 0.05, the result is still valuable when the study that causes high heterogeneity is excluded, but further research is needed to confirm it. Conventional rehabilitation programs mainly employ isometric and isotonic exercises to improve muscle strength, while proprioceptive training programs include balance enhanced training, plyometric stretch-contraction cycles, knee dynamic stability, proprioception, and agility exercises.[36] Progressive training of proprioceptive training programs can stimulate stretch-contraction and enhance the improvement in muscle strength, which may explain the greater muscle strength observed in the proprioceptive training group.[36 37] In addition, failure to improve knee function significantly in this study might be explained by poor transfer of the trained skills to gait. No exercises in the control group resemble the heel toe action of gait, and this degree of task specificity may be required to modify gait. Moreover, the setting of a blank control group and lack of assessment of compliance for two groups may further explain the less functional improvement observed in the proprioceptive training group.[27 36]

In clinic, proprioceptive training is often added into conventional rehabilitation programs to improve the proprioception, muscle strength and function of patients after APM. Although the results of this study show no significant improvement for

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proprioceptive training group in function score, proprioceptive training is still recommended in most studies to rehabilitation programs.[1 25-28 36] In addition, proprioceptive training has been wide used in clinic, such as anterior cruciate ligament reconstruction, knee replacement, knee osteoarthritis, and have significant improvement in terms of the proprioception, rang of motion, balance, muscle strength, knee function, which shows that proprioception training is valuable for patients with knee disease to promote functional recovery.

This study also has some shortcomings. First, as seen in Figure 2, most of the included studies in this study had methodological flaws and small sample size, mostly the lack of participant, therapist and assessor blinding. These factors might overestimate the efficacy of proprioceptive training.[46] More high-quality studies with larger sample should be implemented to confirm the efficacy of proprioceptive training. Second, the start time of intervention and assessment in the included studies was inconsistent. Study showed that the incidence of knee osteoarthritis was approximately 50% within 10 to 20 years from APM.[14] This factor might affect the recovery of knee function and proprioception. In addition, this study generally focused on the effects of proprioceptive training after training and thus lacked studies assessing mid and long-term efficacy. The impact of follow-up time on efficacy of proprioceptive training needs to be explored further. Third, few studies were searched according the inclusion criteria, and different outcomes were also used. Therefore, the conclusions were not supported by sufficient data. More studies are needed to verify the conclusions of this study. Fourth, as seen in Table 1 and Figure 5, two studies[27 37] had blank control groups that were different from those in other studies, which could possibly explain the heterogeneity observed in some outcomes. Due to the small number of studies, further subgroup analysis is not applicable. Future studies should standardize interventions in the control group. Finally, a lack of standardized exercise prescription needed to be noted, such as dose, time, intensity, and type of exercises, which would be the reason of high heterogeneity. Future studies should provide a standard and clear exercise prescription, which could help developing treatment and evaluation methods.

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CONCLUSIONS

Based on low quality evidence, proprioceptive training, combined with conventional rehabilitation training, was significantly effective than conventional rehabilitation training in improving proprioception and the flexor-extensor muscle strength of the knee joint in patients with APM. However, no significant difference had been found between the two groups in improving function of the knee. Given the limited number of included studies, a multicenter randomized controlled trial based on a large sample and a scientific methodology is still needed to further discuss the effects of proprioceptive training on functional recovery in patients after AMP.

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Author Contributions

X.X.L., J.M., D.Z., T.Y.Z., J.W. and S.J. were responsible for conception and design and for the search procedure and check. X.X.L., J.M. and J.W. were responsible for extracting data and quality assessment. In addition, J.M. and H.M.L made great contributions in developing the retrieval strategy, GRADE grading, revising background and discussion section. All authors made substantial contributions to conception and design and interpretation of the data, drafted the manuscript and gave final approval of the final version.

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First author(Year)	Age（T/C）	Sample size（T/C）	Intervention（T/C）	Intervention time from surgery	Frequency	Duration	Outcomes
Yuzhou Li 2014	32.0±6.25 26.5±6.75	12/13	CT+PT/CT	first day	2 times per day, 5-7 times per week	8 weeks	Position sense; Isometric strength
Xiaoling Xiong 2018	29.2±5.12 28.3±5.36	15/15	CT+PT/CT	first day	No description	8 weeks	Isometric strength
Weiyan Huang 2018	18-40	30/30	CT+BT/CT	first day	No description	12 weeks	Lysholm; Position sense
Jianjiang Ouyang 2019	49.20±7.54 48.50±9.93	28/28	CT+PT/CT	6 hours	1 times per day, 30 mins for each	12 weeks	Position sense; Isometric strength
Wenjie Yu 2020	42.22±4.35 42.69±5.08	43/43	CT+PT/CT	first day	2 times per day, 5-7 times per week	8 weeks	Lysholm
Chendi Jiang 2021	40.85±5.47 40.53±5.46	24/24	CT+PT/CT	first day	1 times per day, 6 times per week	8 weeks	Lysholm
Xiaohui Zhang 2017	23.16±3.45 23.25±3.52	15/15	NT/CT	first day	30 mins for each, 3 times per week	8 weeks	Lysholm; Isokinetic strength
Ericsson 2008	45.4±3.2 45.9±3.2	28/28	NT/Blank control	12 months	5 times per week	8 weeks	Isokinetic strength
Hall 2015	42.8±5.4 43.2±5.6	31/31	NT/Blank control	3 months	30-45 mins for each, 1-3 times per week	12 weeks	KOOS; Isokinetic strength

Table 1: Characteristics of included studies. CT: conventional training; PT: proprioceptive training, NT: neuromuscular training, BT: balance training, KOOS: knee injury and osteoarthritis outcome score.

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Study	1	2	3	4	5	6	7	8	9	10	11	Total score	Level
Yuzhou Li	×	×	×	×	×	×	×	√	√	√	√	4	Low
Xiaoling Xiong	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Weiyan Hung	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Jianjiang Ouyang	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Wenjie Yu	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Chendi Jiang	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Xiaohui Zhang	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Ericsson	√	√	√	√	×	×	√	√	√	√	√	8	High
Hall	√	√	√	√	×	×	√	√	√	√	√	8	High

Table 2: Evaluation of the quality of the included documents through PEDro. 1= inclusion exclusion criteria; 2 = randomized group; 3 = allocation concealment; 4 = similar baseline; 5 = subject blinding; 6 = therapist blinding; 7 = assessor blinding; 8 = more than 85% of patient measures; 9 = intention to treat; 10 = between-group analysis; 11 = at least one point measure.√: yes, no risk; ×: no, risky.

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Quality assessment							Number of patients		Effect		Quality	Importance
Number of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	PT	CT	Relative (95% CI)	Absolute (95% CI)		
Proprioceptive test												
	RCT	Very serious*	Very serious†	Not serious	serious•	None	70	71	Risk ratio -1.73 (-2.89 to -0.48)	-	⊕OOO VERY LOW	CRITICAL
Flexor muscle strength of knee												
	RCT	Very serious*	Serious†	serious§	serious•	None	117	117	Risk ratio 0.56 (0.01 to 1.11)	-	⊕OOO VERY LOW	CRITICAL
Extensor muscle strength of knee												
	RCT	Very serious*	Not serious	serious§	serious•	None	129	130	Risk ratio 0.31 (0.06 to 0.56)	-	⊕OOO VERY LOW	CRITICAL
Function scores of knee												
	RCT	Very serious*	Serious†	serious§	serious•▪	None	143	143	Risk ratio 0.85 (-0.13 to 1.84)	-	⊕OOO VERY LOW	CRITICAL

Table 3: GRADE evidence profile.

*We downgraded for a quite large risk of bias, due to lack of randomization, blinding, allocation concealment.

†We downgraded for the high I^2 and reversed the results of functional score.

§We downgraded for the inconsistence of intervention in the control group.

•We downgraded for less than 400 participants.

•We downgraded for no significant benefit.

GRADE, Grading of Recommendations Assessment, Development and Evaluation; PT, proprioceptive training; CT, conventional training; RCT, randomised controlled trial.

Figure 1: Flow chart of literature selection

Figure 2: Meta-analysis of proprioceptive tests(a- all studies; b- after sensitivity analysis)

Figure 3: Meta-analysis of flexor muscle strength of knee(a- all studies; b- after sensitivity analysis)

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Figure 4: Meta-analysis of extensor muscle strength of knee

Figure 5: Meta-analysis of function scores of knee(a- all studies; b- after sensitivity analysis)

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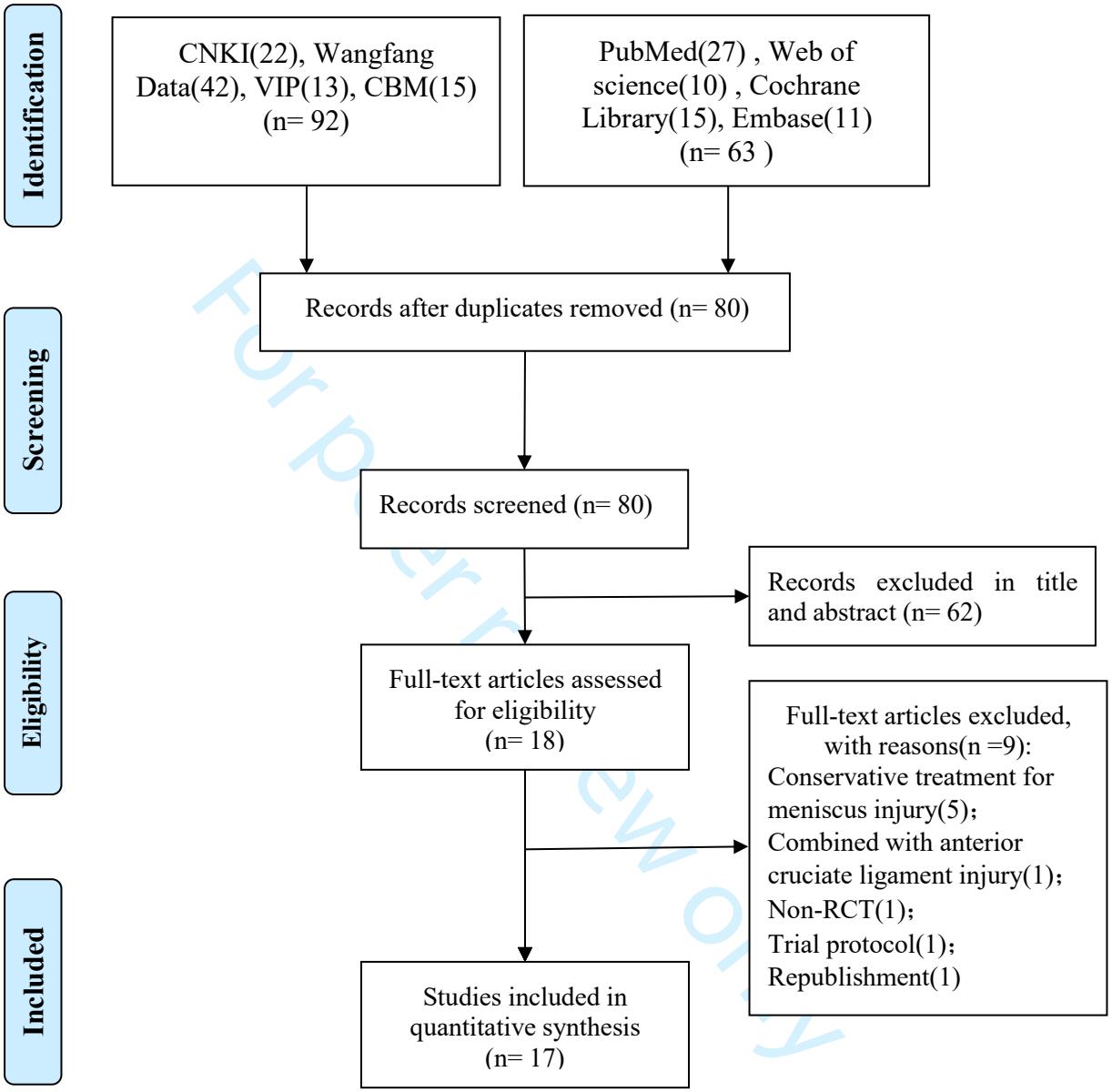


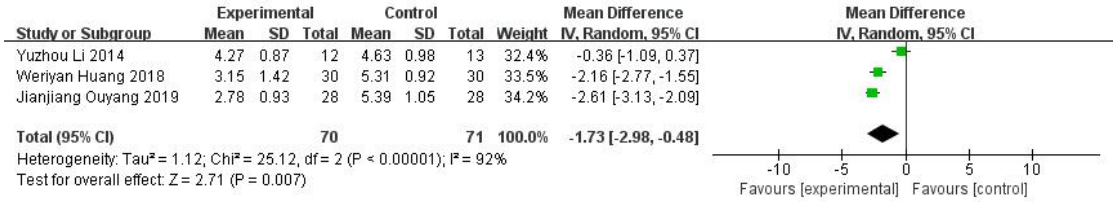
Figure 1: Flow chart of literature selection.

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). *P*referred *R*eporting *I*tems for *S*ystematic Reviews and *M*eta-*A*nalyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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a



b

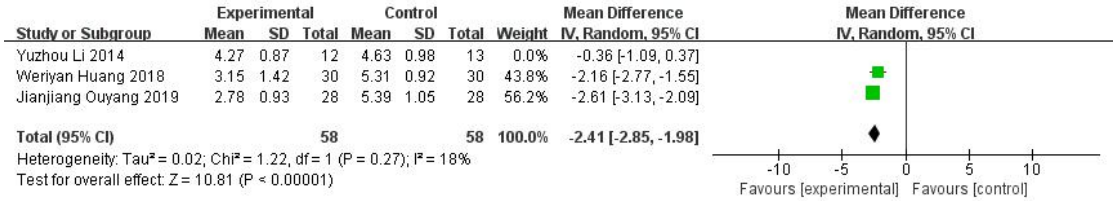
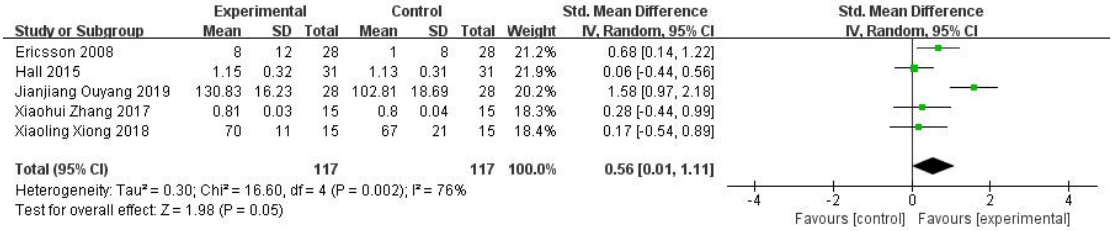


Figure 2: Meta-analysis of proprioceptive tests(a- all studies; b- after sensitivity analysis)

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a



b

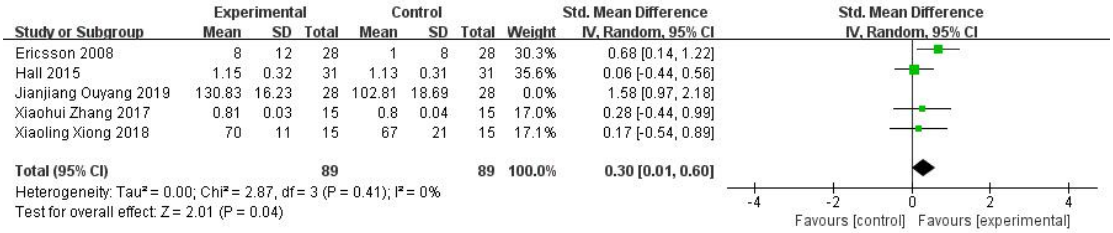


Figure 3: Meta-analysis of flexor muscle strength of knee(a- all studies; b- after sensitivity analysis)

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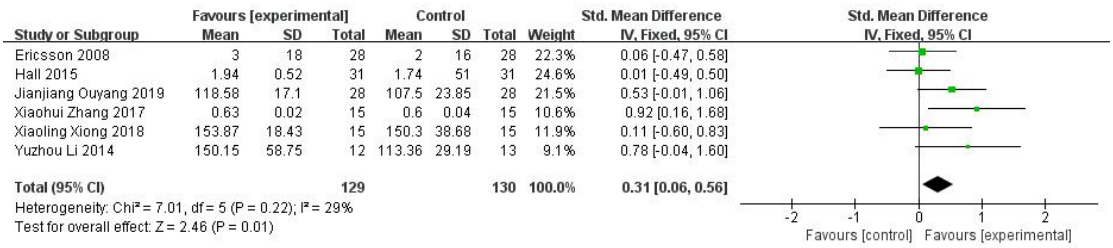
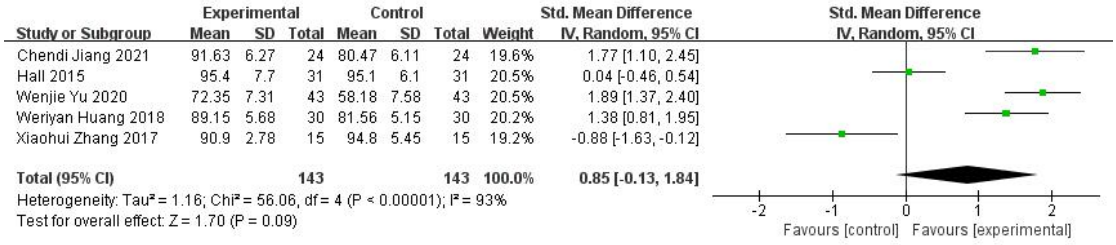


Figure 4: Meta-analysis of extensor muscle strength of knee

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a



b

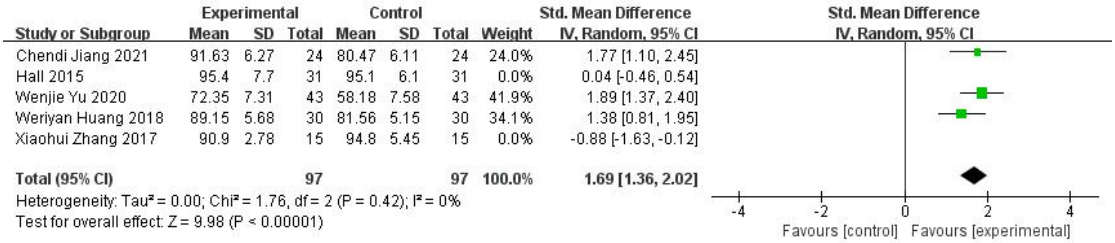


Figure 5: Meta-analysis of function scores of knee(a- all studies; b- after sensitivity analysis)

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Supplemental file
Detailed search strategies for databases

Search strategy for PubMed

#1 Proprioception [Mesh]

#2 Propriocepti* [Title/Abstract] OR Position Sense* [Title/Abstract] OR Sense, Position [Title/Abstract] OR Senses, Position[Title/Abstract] OR Sense of Position [Title/Abstract]

#3 Sensorimotor [Title/Abstract]

#4 Proprioceptive neuromuscular facilitation [Title/Abstract]

#5 Neuromuscular [Title/Abstract]

#6 Balance [Title/Abstract]

#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6

#8 Training [Title/Abstract]

#9 #7 AND #8

#10 Meniscectomy [Mesh]

#11 Meniscectomies [Title/Abstract] OR Arthroscopic Meniscectom* [Title/Abstract] OR Meniscectom*, Arthroscopic [Title/Abstract] OR Meniscal Resection* [Title/Abstract] OR Resection*, Meniscal [Title/Abstract]

#12 #10 OR #11

#13 #9 AND #12

Search strategy for EMBASE

#1 Proprioception /exp

#2 'Proprioception':ti,ab,kw

#3 'Proprioceptive':ti,ab,kw

#4 'Position Sense':ti,ab,kw

#5 'Position Senses':ti,ab,kw

#6 'Sense, Position':ti,ab,kw

#7 'Senses, Position':ti,ab,kw

#8 'Sense of Position':ti,ab,kw

#9 'Sensorimotor':ti,ab,kw

#10 'Proprioceptive neuromuscular facilitation':ti,ab,kw

#11 'Neuromuscular':ti,ab,kw

#12 'Balance':ti,ab,kw

#13 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12

#14 'Training':ti,ab,kw

#15 #13 AND #14

#16 Meniscectomy /exp

#17 'Meniscectomies':ti,ab,kw

#18 'Arthroscopic Meniscectomy':ti,ab,kw

#19 'Arthroscopic Meniscectomies':ti,ab,kw

#20 'Meniscectomy, Arthroscopic':ti,ab,kw

#21 'Meniscectomies, Arthroscopic':ti,ab,kw

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#22 'Meniscal Resection':ti,ab,kw
#23 'Resection, Meniscal':ti,ab,kw
#24 #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23
#25 #15 AND #24

Search strategy for The Cochrane Library

#1 [Proprioception] explode all trees
#2 ("Propriocepti*"):ti,ab,kw OR ("Position Sense*"):ti,ab,kw OR ("Sense, Position"):ti,ab,kw OR ("Senses, Position"):ti,ab,kw OR ("Sense of Position"):ti,ab,kw
#3 ("Sensorimotor"):ti,ab,kw
#4 ("Proprioceptive neuromuscular facilitation"):ti,ab,kw
#5 ("Neuromuscular"):ti,ab,kw
#6 ("Balance"):ti,ab,kw
#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6
#8 ("Training"):ti,ab,kw
#9 #7 AND #8
#10 [Meniscectomy] explode all trees
#11 ("Meniscectomies"):ti,ab,kw OR ("Arthroscopic Meniscectom*"):ti,ab,kw OR ("Meniscectom*, Arthroscopic"):ti,ab,kw OR ("Meniscal Resection*"):ti,ab,kw OR ("Resection*, Meniscal"):ti,ab,kw
#12 #10 OR #11
#13 #9 AND #12

Search strategy for Web of science

#1 TS=(Propriocept* OR Position Sense* OR Sense*, Position OR Sense of Position OR Sensorimotor OR Proprioceptive neuromuscular facilitation OR Neuromuscular OR Balance)
#2 TS= (Training)
#3 TS=(Meniscectom* OR Arthroscopic Meniscectom* OR Meniscectom*, Arthroscopic OR Meniscal Resection* OR Resection*, Meniscal)
#4 (#1 AND #2 AND #3)

Search strategy for China National Knowledge Infrastructure (CNKI)

#1 "本体感觉"[主题]
#2 "平衡"[主题]
#3 "神经肌肉"[主题]
#4 "本体感觉神经肌肉促进技术"[主题]
#5 "感觉运动"[主题]
#6 "训练"[主题]
#7 "半月板术后"[主题]
#8 "半月板切除术"[主题]
#9 "半月板成型术"[主题]
#10 "半月板修整术"[主题]
#11 #1 OR #2 OR #3 OR #4 OR #1
#12 #7 OR #8 OR #9 OR #10

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#13 #6 AND #11 AND #12

Search strategy for Technology Periodical Database (VIP)

#1 “本体感觉”[题名或关键词]
#2 “平衡”[题名或关键词]
#3 “神经肌肉”[题名或关键词]
#4 “本体感觉神经肌肉促进技术”[题名或关键词]
#5 “感觉运动”[题名或关键词]
#6 “训练”[题名或关键词]
#7 “半月板术后”[题名或关键词]
#8 “半月板切除术”[题名或关键词]
#9 “半月板成型术”[题名或关键词]
#10 “半月板修整术”[题名或关键词]
#11 #1 OR #2 OR #3 OR #4 OR #1
#12 #7 OR #8 OR #9 OR #10
#13 #6 AND #11 AND #12

Search strategy for WanFang Data

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#2 “平衡”[主题]
#3 “神经肌肉”[主题]
#4 “本体感觉神经肌肉促进技术”[主题]
#5 “感觉运动”[主题]
#6 “训练”[主题]
#7 “半月板术后”[主题]
#8 “半月板切除术”[主题]
#9 “半月板成型术”[主题]
#10 “半月板修整术”[主题]
#11 #1 OR #2 OR #3 OR #4 OR #1
#12 #7 OR #8 OR #9 OR #10
#13 #6 AND #11 AND #12

Search strategy for China Biology Medicine (CBM)

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#2 “平衡”[常用字段:智能]
#3 “神经肌肉”[常用字段:智能]
#4 “本体感觉神经肌肉促进技术”[常用字段:智能]
#5 “感觉运动”[常用字段:智能]
#6 “训练”[常用字段:智能]
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#8 “半月板切除术”[常用字段:智能]
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#12 #7 OR #8 OR #9 OR #10
#13 #6 AND #11 AND #12

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	3-4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	5-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	6,7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	7
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	7
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	6,7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	7
Study characteristics	17	Cite each included study and present its characteristics.	8
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	8
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	8
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	8-9
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	8-9
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	8-9
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	8-9
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	-
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	9-10
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	10-12
	23b	Discuss any limitations of the evidence included in the review.	12-13
	23c	Discuss any limitations of the review processes used.	12-13
	23d	Discuss implications of the results for practice, policy, and future research.	13
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	4-5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	4-5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	4-5
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	13
Competing interests	26	Declare any competing interests of review authors.	14
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	14

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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BMJ Open

Effects of proprioceptive training in the recovery of patients submitted to meniscus surgery: systematic review and Meta-analysis

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Manuscript ID	bmjopen-2021-055810.R2
Article Type:	Original research
Date Submitted by the Author:	23-Mar-2022
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Primary Subject Heading:	Rehabilitation medicine
Secondary Subject Heading:	Rehabilitation medicine, Sports and exercise medicine
Keywords:	Rehabilitation medicine < INTERNAL MEDICINE, ORTHOPAEDIC & TRAUMA SURGERY, Knee < ORTHOPAEDIC & TRAUMA SURGERY, Orthopaedic sports trauma < ORTHOPAEDIC & TRAUMA SURGERY

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Effects of proprioceptive training in the recovery of patients submitted to meniscus surgery: systematic review and Meta-analysis

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JM, XXL, HML are the co-first authors and contributed equally.

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ABSTRACT

Objective To evaluate the effects of proprioceptive training on rehabilitation of knee after arthroscopic partial meniscectomy.

Design PubMed, EMBASE, The Cochrane Library, Web of Science, China National Knowledge Infrastructure (CNKI), Technology Periodical Database (VIP), WanFang Data, and China Biology Medicine (CBM) were searched until December 2021 for randomized controlled trials.

Participants Patients who have undergone arthroscopic partial meniscectomy for meniscus injury caused by traumatic tear.

Results A total of 9 studies with 453 patients were included in this study for meta-analysis, and 2/9 with high quality, 6/9 with moderate quality. Based on very low quality evidence, the pooled effect showed significant improvement for proprioceptive training group in proprioception test ($P< 0.05$, $I^2= 18\%$), knee extensor muscle strength ($P< 0.05$, $I^2= 29\%$), knee flexor muscle strength ($P< 0.05$, $I^2= 0\%$) and knee function score ($P< 0.05$, $I^2= 0\%$) compared to conventional training group in patients after arthroscopic partial meniscectomy.

Conclusion: Based on very low quality, adding proprioceptive training to conventional rehabilitation programs might be beneficial to promote functional recovery for patients after arthroscopic partial meniscectomy. It is necessary to carry out more samples and higher quality large-scale studies to provide high evidence in the future.

PROSPERO registration number CRD42020213201

Strengths and limitations of this study

- ► This meta-analysis was preregistered and conducted in line with the Preferred Reporting Items for Systematic Reviews and Meta-analyses guidelines.
- ► Eight electronic databases were searched to provide a comprehensive range of studies.
- ► Differences in the control interventions, the time-course of treatment, the start time of intervention and assessment may increase heterogeneity and bias.

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INTRODUCTION

Meniscal injuries are one of the most common diseases in orthopedics, with an annual average of 60- 70 meniscal injuries per 100, 000 cases concerning knee injuries.[1 2] Meniscus tears were detected in up to 80% of cases when people have magnetic resonance imaging scans to check the knee.[3] This type of injury is generally classified into two etiologies.[4-6] One is caused by traumatic or sports injuries when axial loads are transmitted directly to the flexed knee joint with rotation. The other is defined as a degenerative change and often accompanied by degenerative changes in cartilage, mostly found in middle-aged and elderly people, which is often involved in knee osteoarthritis.[1 3] Arthroscopic partial meniscectomy (APM) is among the most common procedures performed for their treatment.[1 7 8] Study shows that about 636 000 knee arthroscopy procedures are performed every year in the United States.[3] In addition, recent studies have shown that surgical treatment has been transferred toward facilitating meniscal repair to maintain meniscal tissue integrity whenever possible, prevent secondary cartilage degeneration and improve knee joint function.[1 9-11] However, postoperative patients experience pain and swelling leading to loss of range of movement, proprioceptive deficits, neuromuscular and biomechanical changes, decreased quadriceps femoris muscle strength, and further leading to dysfunction. The dysfunction especially in proprioception which perceives changes in the state of motion and initiates protection and muscular reflexes to keep dynamic stabilizing are prone to increase the risk of degenerative disease or reinjury in the long-term after surgery.[1 12-14]

Proprioception plays an important role in postoperative rehabilitation. The inputs from the sensory organs are processed in the brain and integrated with visual and vestibular information to generate a sense of position and movement through space. Proprioceptive training can stimulate the sensory organ to send specific signals to control the relevant muscles to maintain stability through specific dynamic movement exercises such as practices of balancing, positioning, gait flexibility, agility, and

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neuromuscular.[15-17] In recent years, adding proprioceptive training to conventional rehabilitation has been widely used in postoperative rehabilitation of joints at different times, some on the first day after surgery, and some three months after surgery, which proved to be effective in improving proprioception and activity function.[1] Besides, benefits could be obtained after proprioceptive training for not only patients with anterior cruciate ligament reconstruction,[18 19] knee osteoarthritis,[20] hip and knee replacement,[16 21] but also athletes with the prevention of sports injuries.[22-24]

Proprioception training is beneficial to patients with APM. According to previous trials, some studies have shown that proprioceptive training has a significant improvement on knee rehabilitation for patients with APM.[25 26] However, no significant difference is shown between patients undergoing proprioceptive training and counterparts without such treatment after surgery.[27 28] Considering the differences in the efficacy of proprioceptive training applied in the rehabilitation, this study will not only meta-analyze the effect of proprioceptive training on the recovery of knee proprioception and muscle strength among patients with APM caused by traumatic tear but also provide reliable medical evidence for the efficacy of proprioceptive training on the recovery of proprioception and function after APM.

METHODS

Protocol and registration

This systematic review and meta-analysis was conducted following the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-analyses(PRISMA). The protocol of the current review was registered on the International Prospective Register of Systematic Reviews(PROSPERO, 2021, CRD42020213201).

Patient and public involvement

Patients and/ or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

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Search strategy

A comprehensive search using the electronic databases of PubMed, EMBASE, Cochrane Library, Web of Science, CNKI, VIP, WangFang Data, and CBM was conducted in December 2021. Following keywords and their varies were used: proprioceptive training, sensorimotor training, proprioceptive neuromuscular facilitation, neuromuscular training, balance training, meniscectomy. The language was restricted to Chinese and English. Detailed search strategies based on guidance from the Cochrane handbook for all the above databases were shown in the supplemental appendix S1.

Study selection

All records were managed with Endnote X9. The republished articles and no English abstract articles were excluded. Two authors (XX and MJ) screened the studies and extracted the data independently according to the inclusion and exclusion criteria. The information of author, publication year, demographics of participants, intervention, the start time of intervention, training time, and outcomes were recorded. Any disagreements were resolved by discussion or umpired with a third reviewer (WJ). After we reviewed relevant articles, eligibility criteria for this review based on PICOS frameworks (Population, Intervention, Comparison, Outcome and Study) were as follows: (1) Population: patients who have undergone APM for meniscus injury caused by traumatic tear, whose race, nationality, and duration of disease are not limited, patients who have undergone meniscal surgery combined with other procedures such anterior cruciate ligament reconstruction would be excluded; (2) Interventions: proprioceptive training alone or combined training with conventional rehabilitation were seemed as experimental groups; (3) Comparison: conventional rehabilitation training (including ankle pump movement, continuous passive motion, weight-bearing exercise, strength training, etc.) were applied to control groups, control groups intervention did not include proprioceptive training; (4) Outcome: at least one of the outcomes of extensor, flexor muscle strength, knee function score or

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proprioception test was reported; (5) Study: randomized controlled trial (RCT) published in English or Chinese.

Risk of bias assessment

Two investigators independently assessed the quality of the literature using the PEDro scale.[29] Eleven criteria were used in the PEDro scale, and each criterion was rated as "yes" or "no". Each yes would earn one score. However, the first question, “Whether we should make a detailed sampling criteria for the experiment” is excluded. The full mark is 10. A total PEDro score of ≥ 7 was considered as high quality, 5- 6 as moderate quality, and≤ 4 as low quality.[29] The scores were given independently by two researchers, and results would be rediscussed with a third researcher in case of disagreement.

Data extraction

Two researchers independently selected the literature and extracted the data according to sampling criteria. The selection of the literature was based upon the relevance of the research topic. Irrelevant studies would be excluded by reading the title and abstract of the literature. Final decision would be made through a detailed reading of the full text. Data extraction was performed independently by two researchers, including: first author, year, age, sample size, type, time and frequency of intervention, and outcome indicators. Disagreements would be resolved through discussions.

Statistical analysis

Data included in the study were analyzed using Review Manager 5.3. Mean Difference (MD) was used for consistent measuring units. Standardized Mean Difference (SMD) was used for inconsistent measuring units to calculate the effect size and its 95% confidence interval (CI) for each combination.[30] The heterogeneity of treatment effects was examined by calculating the I^2 index. Interpretation of I^2 is as follows: 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to

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90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity. When I^2 was less than 50%, the combined effect was considered as mild heterogeneity and a fixed-effect model was utilized for meta-analysis; when I^2 was greater than 50%, the heterogeneity was considered to be high. Therefore, a random-effect model was used for meta-analysis, and sensitivity analysis was performed to identify the source of heterogeneity. The absence of blinding and differences in the control interventions, the time-course of treatment, the start time of intervention, and assessment may increase heterogeneity. *P* value level of 0.05 was set for between-group differences.

GRADE Assessment

The GRADE Approach (Grading of Recommendations, Assessment, Development, and Evaluations) was used to appraise and summarise the body of evidence.[31] GRADE is an internationally approved standard for managing complex evidence reviews. For results based only on RCTs, certainty was initially considered as high. Thereafter, certainty could be rated down based on factors such as the risk of bias, imprecision, inconsistency, indirectness, and potential publication bias.[31 32]

RESULTS

Study characteristics

As of December 2021, 155 potentially relevant studies were retrieved according to the sampling criteria, and 9 relevant articles were selected, 6 in Chinese[26-28 33-35] and 3 in English[27 36 37], with a total of 453 patients. A detailed selection process is shown in Figure 1.

All included literature entailed RCTs concerning the effect of proprioceptive training on knee function among patients with APM.[25-28 33-37] Seven studies reported that the patients combined with ACL injuries were excluded,[25 27 33-37] but the others[26 28] did not report in the methods. There are also various methods for assessing knee flexor and extensor muscle strength,[38 39] such as isokinetic strength,[27 36] isometric strength,[28 33]

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peak torque strength,[37] and relative torque strength.[26] There are also various methods for assessing knee function scores such as Lysholm scores[25 34-36] and Knee Injury and Osteoarthritis Outcome Score.[27] Specific study characteristics are shown in Table 1.

Quality assessment

The quality of the included studies was evaluated according to the PEDro quality assessment scale, most of all had methodological flaws in the subject, therapist, and assessor blinding. Two studies obtained high quality,[27 37] six studies obtained moderate quality,[25 26 33-36] and one study obtained low quality,[28] as detailed in Table 2.

Effects of proprioceptive training

Proprioceptive test

Three studies,[25 28 33] assessing 141 participants, performed knee proprioceptive test. All three studies assessed positioning sense using the threshold to detection of passive motion, and MD was applied in the process of data merging. Pooling of the data using a random effects model (I^2= 92%, Figure 2-a) showed a statistical significance compared with control group (MD= -1.73, 95% CI= [-2.98, -0.48], P< 0.001), which could prove the positive effects brought by proprioceptive training for patients' knee proprioception. After sensitivity analysis, the study[28] was identified as a source of high heterogeneity, which decreased significantly (P< 0.00001, I^2= 18%, Figure 2-b) after being excluded. It was considered as a result of the small sample size and improper methodology used in the study.

Flexor muscle strength

Five studies,[26 27 33 36 37] assessing 234 participants, performed the knee flexor muscle strength test. Pooling of the data using a random effects model(I^2= 76%, Figure 3-a) showed no statistical significance compared with control group (SMD= 0.56, 95% CI= [0.01, 1.11], P= 0.05). After sensitivity analysis, the study[33] was identified as the source of high heterogeneity, which decreased significantly (I^2= 0%, P= 0.04, Figure 3-b) after

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being excluded, which could justify the positive effects on knee flexor (hamstring) muscle strength brought by proprioceptive training. It was considered as a result of the seniority of the included patients and improper methodology in the study.

Extensor muscle strength

Six studies,[26-28 33 36 37] assessing 259 participants, performed the knee extensor muscle strength test. Pooling of the data using a fixed effects model(I^2= 29%, Figure 4) showed a statistical significance compared with control group (SMD= 0.31, 95% CI= [0.06, 0.56], *P*= 0.01), which could justify that proprioceptive training could improve knee extensor (quadriceps) muscle strength.

Function scores

Five studies, assessing 286 participants, performed the knee function scores assessed by Lysholm score[27] and knee injury and osteoarthritis outcome score (KOOS)[25 34-36]. The. Pooling of the data using a random effects model(I^2= 93%, Figure 5-a) showed no significant difference between the two groups (SMD= 0.85, 95% CI= [-0.13, 1.84], *P*= 0.09). After sensitivity analysis, two studies[27 36] were found to be the source of high heterogeneity, which decreased significantly (I^2= 0%, *P*< 0.00001, Figure 5-b) after being excluded one by one. It was considered partly because of the blank control as the control group.

GRADE approach level of evidence

Pooled results of the proprioceptive test, knee flexor muscle strength, knee extensor muscle strength, knee function scores comparing proprioceptive training group to conventional training group were considered of very low quality. See details of the Grade Approach and conclusions in Table 3.

DISCUSSION

This study is the first systematic review and meta-analysis to evaluate the functional

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effects of proprioceptive training after arthroscopic partial meniscectomy. Nine RCTs with 453 patients containing 2 high-quality studies, 6 moderate-quality studies, and 1 low-quality study respectively were included. According to the results of meta-analysis, the included studies showed that the proprioceptive training could significantly improve knee proprioception, flexion and extension muscle strength in patients after arthroscopic partial meniscectomy, while the positive effects on knee function scores were not significant. GRADE approach showed that all the pooled results comparing the proprioceptive training group to the conventional training group were considered of very low quality.

Proprioceptive training could significantly improve knee proprioception in patients after APM. The previous study[40] showed that patients with isolated meniscal tears had a significant proprioceptive deficit when compared to their uninjured contra-lateral knee and the healthy subjects. Located in the outer third place of the meniscus, the proprioceptors could sense the condition of movement from the knee joints and their surrounding muscles, playing an important role in the maintenance of knee stability by regulating the amount of knee muscle strength through proprioceptive feedback.[15 25 28] Proprioception refers to the signal of body changes generated by proprioceptors in static or dynamic motion to specific parts of the body, which generally include the sense of static position, dynamic motion velocity, the direction of acceleration and the perception of joint pressure.[41] Proprioceptive feedback mechanism is a subjective perception of knee stability.[42] Once the knee proprioception is stimulated, signals are sent to the central nervous system, including the spinal cord, brainstem, and cerebral cortex, where they are analyzed and transmitted downward to protect the knee from injury by stimulating or inhibiting the corresponding muscles, adjusting posture and balancing in time.[19 43] After the meniscus injury partially or completely, the proprioception and neuromuscular control ability of the knee are significantly decreased.[28 43] Partial meniscus impairment could also be found after arthroscopic partial meniscectomy. Arthroscopic partial meniscectomy could not result in a significant improvement in proprioceptive function.[44] Impaired proprioceptive

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feedback mechanism would predispose to reflex joint instability and irregular postural reflexes, leading to an increasing risk of degenerative disease or re-injury.[19 36 45] Postoperative proprioceptive training could effectively restore the proprioceptive function, which is consistent with our results.

The results in this study show significant improvement for the proprioceptive training group in knee extensor muscle strength, but no differences in knee flexor muscle strength and function score compared to the conventional training group. Although the *P* value in improving flexor muscle strength is 0.05, the result is still valuable when the study that causes high heterogeneity is excluded, but further research is needed to confirm it. Conventional rehabilitation programs mainly employ isometric and isotonic exercises to improve muscle strength, while proprioceptive training programs include balance enhanced training, plyometric stretch-contraction cycles, knee dynamic stability, proprioception, and agility exercises.[36] Progressive training of proprioceptive training programs can stimulate stretch-contraction and enhance the improvement in muscle strength, which may explain the greater muscle strength observed in the proprioceptive training group.[36 37] In addition, failure to improve knee function significantly in this study might be explained by the poor transfer of the trained skills to gait. No exercises in the control group resemble the heel-toe action of gait, and this degree of task specificity may be required to modify gait. Moreover, the setting of a blank control group and lack of assessment of compliance for two groups may further explain the less functional improvement observed in the proprioceptive training group.[27 36]

In the clinic, proprioceptive training is often added into conventional rehabilitation programs to improve the proprioception, muscle strength, and function of patients after APM. Although the results of this study show no significant improvement for the proprioceptive training group in function score, proprioceptive training is still recommended in most studies to rehabilitation programs.[1 25-28 36] The findings obtained by this review reinforce what is found in other conditions (knee replacement, ACL reconstruction), suggesting proprioceptive to induce functional benefits.

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This study also has some shortcomings. First, as seen in Figure 2 and Table 3, most of the included studies in this study had methodological flaws and a small sample size, mostly the lack of participant, therapist and assessor blinding. These factors might overestimate the efficacy of proprioceptive training.[46] More high-quality studies with a larger sample should be implemented to confirm the efficacy of proprioceptive training. Second, the start time of intervention and assessment in the included studies was inconsistent. The study showed that the incidence of knee osteoarthritis was approximately 50% within 10 to 20 years from APM.[14] This factor might affect the recovery of knee function and proprioception. In addition, this study generally focused on the effects of proprioceptive training after training and thus lacked studies assessing mid and long-term efficacy. The impact of follow-up time on the efficacy of proprioceptive training needs to be explored further. Third, few studies were searched according to the inclusion criteria, and different outcomes were also used. Therefore, the conclusions were not supported by sufficient data. More studies are needed to verify the conclusions of this study. Fourth, as seen in Table 1 and Figure 5, two studies[27 37] had blank control groups that were different from those in other studies, which could explain the heterogeneity observed in some outcomes. Due to the small number of studies, further subgroup analysis is not applicable. Future studies should standardize interventions in the control group. Fifth, the two studies[26 28] did not explicitly report that patients combined with ACL injuries were excluded, which may lead to potential heterogeneity. None of the original included studies reported whether the included patients had been diagnosed with a traumatic or degenerative meniscal tear before arthroscopic partial meniscectomy. Future studies should develop strict inclusion and exclusion criteria. Finally, a lack of standardized exercise prescription needed to be noted, such as dose, time, intensity, and type of exercises, which would be the reason for high heterogeneity. Future studies should provide a standard and clear exercise prescription, which could help develop treatment and evaluation methods.

CONCLUSIONS

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Based on very low quality, adding proprioceptive training to conventional rehabilitation training may induce benefits for knee flexor-extensor strength and proprioception in patients after arthroscopic partial meniscectomy. Given the limited number of included studies, a multicenter randomized controlled trial based on a large sample and a scientific methodology is still needed to further discuss the effects of proprioceptive training on functional recovery in patients after arthroscopic partial meniscectomy.

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Author Contributions

X.X.L., J.M., D.Z., T.Y.Z., J.W. and S.J. were responsible for conception and design and for the search procedure and check. X.X.L., J.M. and J.W. were responsible for extracting data and quality assessment. In addition, J.M. and H.M.L made great contributions in developing the retrieval strategy, GRADE grading, revising background and discussion section. All authors made substantial contributions to conception and design and interpretation of the data, drafted the manuscript and gave final approval of the final version.

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First author(Year)	Age（T/C）	Sample size（T/C）	Intervention（T/C）	Intervention time from surgery	Frequency	Duration	Outcomes
Yuzhou Li 2014	32.0±6.25 26.5±6.75	12/13	CT+PT/CT	first day	2 times per day, 5-7 times per week	8 weeks	Position sense; Isometric strength
Xiaoling Xiong 2018	29.2±5.12 28.3±5.36	15/15	CT+PT/CT	first day	No description	8 weeks	Isometric strength
Weiyan Huang 2018	18-40	30/30	CT+BT/CT	first day	No description	12 weeks	Lysholm; Position sense
Jianjiang Ouyang 2019	49.20±7.54 48.50±9.93	28/28	CT+PT/CT	6 hours	1 times per day, 30 mins for each	12 weeks	Position sense; Isometric strength
Wenjie Yu 2020	42.22±4.35 42.69±5.08	43/43	CT+PT/CT	first day	2 times per day, 5-7 times per week	8 weeks	Lysholm
Chendi Jiang 2021	40.85±5.47 40.53±5.46	24/24	CT+PT/CT	first day	1 times per day, 6 times per week	8 weeks	Lysholm
Xiaohui Zhang 2017	23.16±3.45 23.25±3.52	15/15	NT/CT	first day	30 mins for each, 3 times per week	8 weeks	Lysholm; Isokinetic strength
Ericsson 2008	45.4±3.2 45.9±3.2	28/28	NT/Blank control	12 months	5 times per week	8 weeks	Isokinetic strength
Hall 2015	42.8±5.4 43.2±5.6	31/31	NT/Blank control	3 months	30-45 mins for each, 1-3 times per week	12 weeks	KOOS; Isokinetic strength

Table 1: Characteristics of included studies. CT: conventional training; PT: proprioceptive training, NT: neuromuscular training, BT: balance training, KOOS: knee injury and osteoarthritis outcome score.

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Study	1	2	3	4	5	6	7	8	9	10	11	Total score	Level
Yuzhou Li	×	×	×	×	×	×	×	√	√	√	√	4	Low
Xiaoling Xiong	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Weiyan Hung	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Jianjiang Ouyang	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Wenjie Yu	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Chendi Jiang	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Xiaohui Zhang	√	√	×	√	×	×	×	√	√	√	√	6	Fair
Ericsson	√	√	√	√	×	×	√	√	√	√	√	8	High
Hall	√	√	√	√	×	×	√	√	√	√	√	8	High

Table 2: Evaluation of the quality of the included documents through PEDro. 1= inclusion exclusion criteria; 2 = randomized group; 3 = allocation concealment; 4 = similar baseline; 5 = subject blinding; 6 = therapist blinding; 7 = assessor blinding; 8 = more than 85% of patient measures; 9 = intention to treat; 10 = between-group analysis; 11 = at least one point measure.√: yes, no risk; ×: no, risky.

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Quality assessment							Number of patients		Effect		Quality	Importance
Number of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	PT	CT	Relative (95% CI)	Absolute (95% CI)		
Proprioceptive test												
	RCT	Very serious*	Very serious†	Not serious	serious•	None	70	71	Risk ratio -1.73 (-2.89 to -0.48)	-	⊕OOO VERY LOW	CRITICAL
Knee flexor muscle strength												
	RCT	Very serious*	Serious†	serious§	serious•	None	117	117	Risk ratio 0.56 (0.01 to 1.11)	-	⊕OOO VERY LOW	CRITICAL
Knee extensor muscle strength												
	RCT	Very serious*	Not serious	serious§	serious•	None	129	130	Risk ratio 0.31 (0.06 to 0.56)	-	⊕OOO VERY LOW	CRITICAL
Knee function scores												
	RCT	Very serious*	Serious†	serious§	serious•▪	None	143	143	Risk ratio 0.85 (-0.13 to 1.84)	-	⊕OOO VERY LOW	CRITICAL

Table 3: GRADE evidence profile.

*We downgraded for a quite large risk of bias, due to lack of randomization, blinding, allocation concealment.

†We downgraded for the high I^2 and reversed the results of functional score.

§We downgraded for the inconsistence of intervention in the control group.

•We downgraded for less than 400 participants.

•We downgraded for no significant benefit.

GRADE, Grading of Recommendations Assessment, Development and Evaluation; PT, proprioceptive training; CT, conventional training; RCT, randomised controlled trial.

Figure 1: Flow chart of literature selection

Figure 2: Meta-analysis of proprioceptive tests(a- all studies; b- after sensitivity analysis)

Figure 3: Meta-analysis of knee flexor muscle strength(a- all studies; b- after sensitivity analysis)

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Figure 4: Meta-analysis of knee extensor muscle strength

Figure 5: Meta-analysis of knee function scores(a- all studies; b- after sensitivity analysis)

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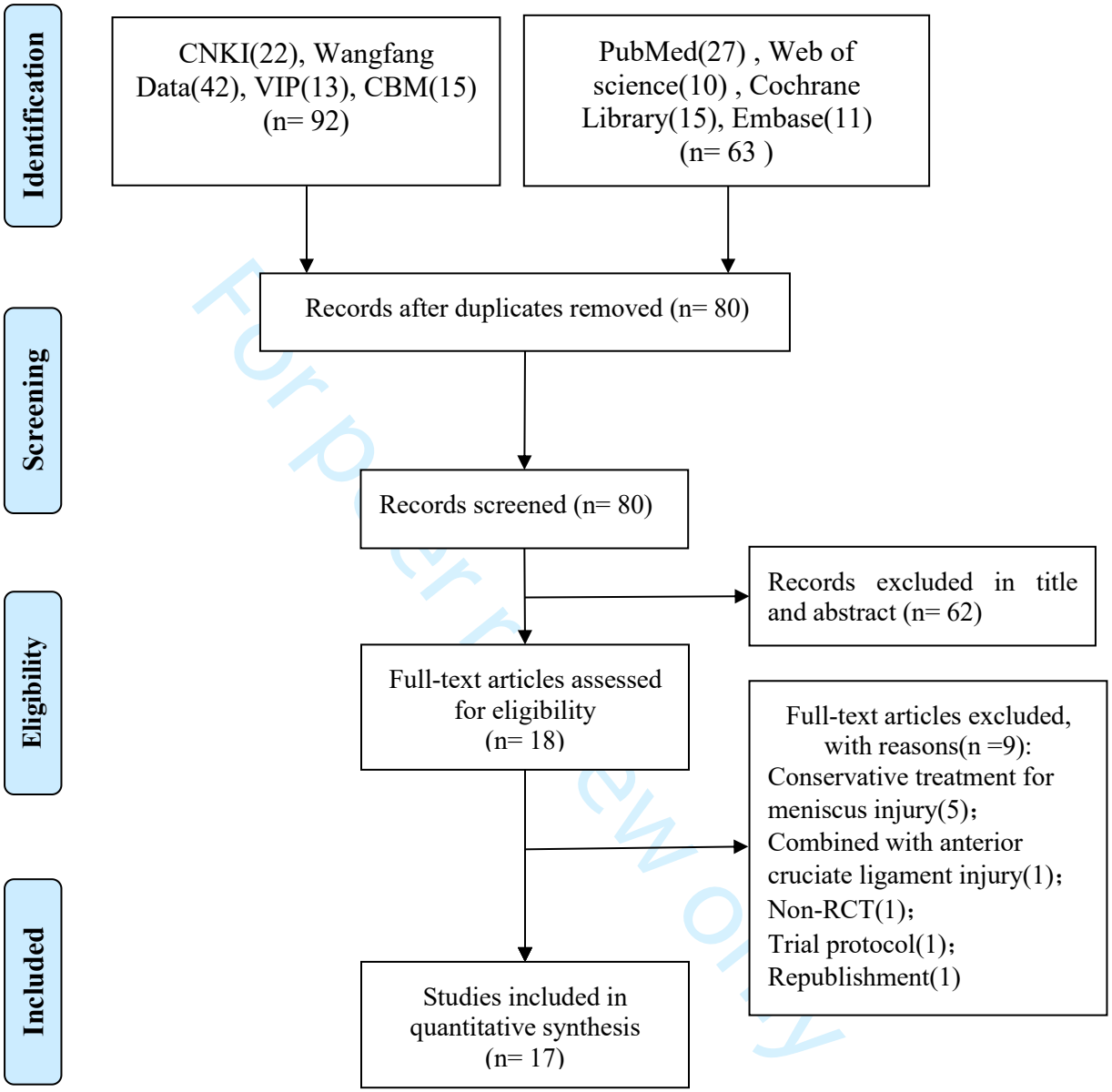


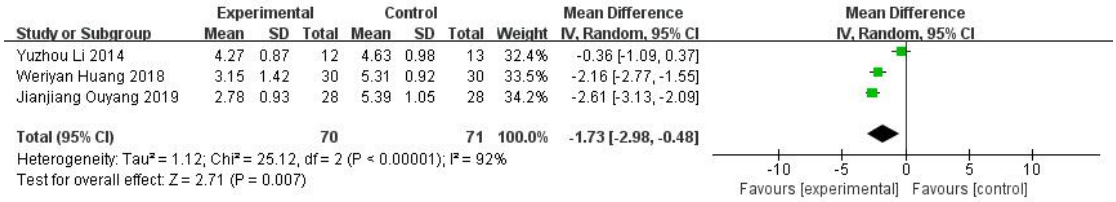
Figure 1: Flow chart of literature selection.

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). *P*referred *R*eporting *I*tems for *S*ystematic Reviews and *M*eta-*A*nalyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit www.prisma-statement.org.

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a



b

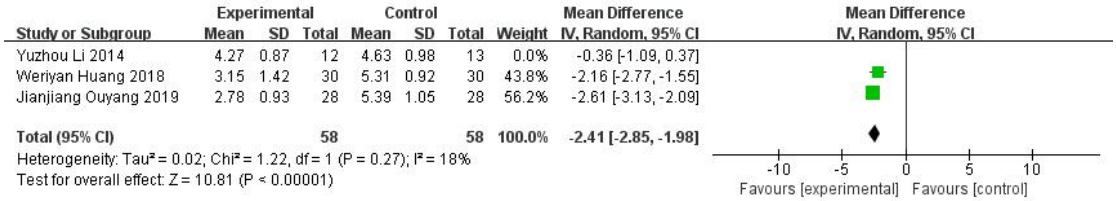
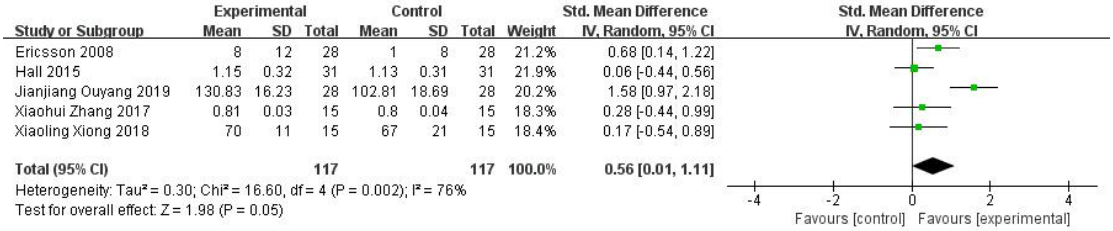


Figure 2: Meta-analysis of proprioceptive tests(a- all studies; b- after sensitivity analysis)

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a



b

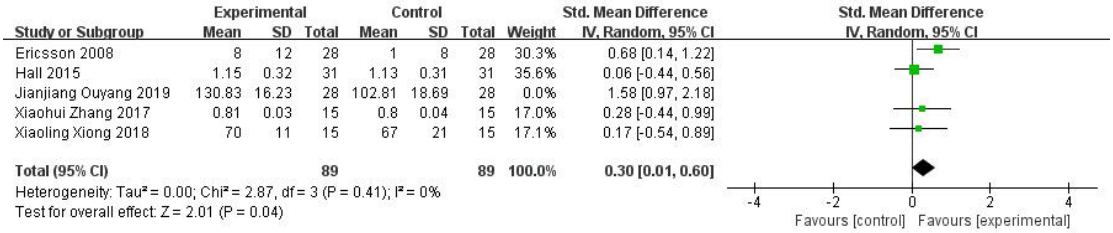


Figure 3: Meta-analysis of the knee flexor muscle strength(a- all studies; b- after sensitivity analysis)

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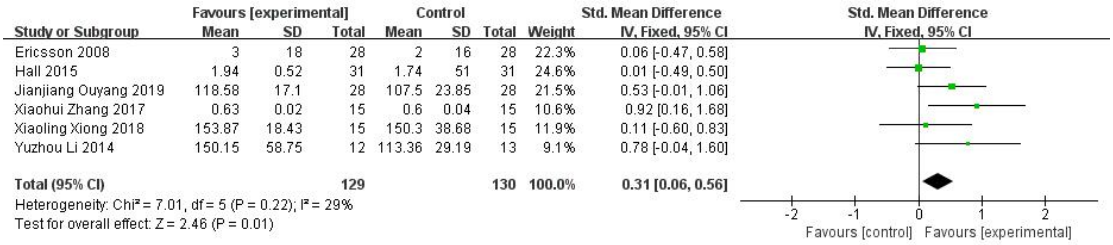
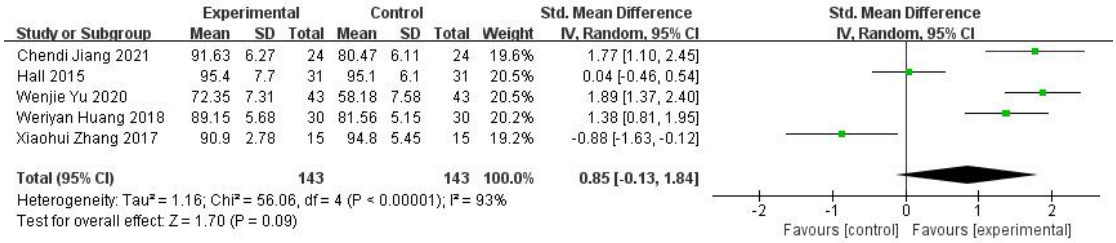


Figure 4: Meta-analysis of the knee extensor muscle strength.

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a



b

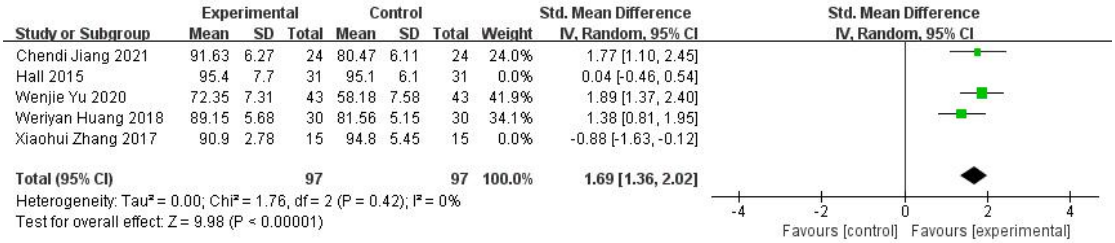


Figure 5: Meta-analysis of the knee function scores(a- all studies; b- after sensitivity analysis)

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Supplemental file
Detailed search strategies for databases

Search strategy for PubMed

#1 Proprioception [Mesh]

#2 Propriocepti* [Title/Abstract] OR Position Sense* [Title/Abstract] OR Sense, Position [Title/Abstract] OR Senses, Position[Title/Abstract] OR Sense of Position [Title/Abstract]

#3 Sensorimotor [Title/Abstract]

#4 Proprioceptive neuromuscular facilitation [Title/Abstract]

#5 Neuromuscular [Title/Abstract]

#6 Balance [Title/Abstract]

#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6

#8 Training [Title/Abstract]

#9 #7 AND #8

#10 Meniscectomy [Mesh]

#11 Meniscectomies [Title/Abstract] OR Arthroscopic Meniscectom* [Title/Abstract] OR Meniscectom*, Arthroscopic [Title/Abstract] OR Meniscal Resection* [Title/Abstract] OR Resection*, Meniscal [Title/Abstract]

#12 #10 OR #11

#13 #9 AND #12

Search strategy for EMBASE

#1 Proprioception /exp

#2 'Proprioception':ti,ab,kw

#3 'Proprioceptive':ti,ab,kw

#4 'Position Sense':ti,ab,kw

#5 'Position Senses':ti,ab,kw

#6 'Sense, Position':ti,ab,kw

#7 'Senses, Position':ti,ab,kw

#8 'Sense of Position':ti,ab,kw

#9 'Sensorimotor':ti,ab,kw

#10 'Proprioceptive neuromuscular facilitation':ti,ab,kw

#11 'Neuromuscular':ti,ab,kw

#12 'Balance':ti,ab,kw

#13 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12

#14 'Training':ti,ab,kw

#15 #13 AND #14

#16 Meniscectomy /exp

#17 'Meniscectomies':ti,ab,kw

#18 'Arthroscopic Meniscectomy':ti,ab,kw

#19 'Arthroscopic Meniscectomies':ti,ab,kw

#20 'Meniscectomy, Arthroscopic':ti,ab,kw

#21 'Meniscectomies, Arthroscopic':ti,ab,kw

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#22 'Meniscal Resection':ti,ab,kw
#23 'Resection, Meniscal':ti,ab,kw
#24 #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23
#25 #15 AND #24

Search strategy for The Cochrane Library

#1 [Proprioception] explode all trees
#2 ("Propriocepti*"):ti,ab,kw OR ("Position Sense*"):ti,ab,kw OR ("Sense, Position"):ti,ab,kw OR ("Senses, Position"):ti,ab,kw OR ("Sense of Position"):ti,ab,kw
#3 ("Sensorimotor"):ti,ab,kw
#4 ("Proprioceptive neuromuscular facilitation"):ti,ab,kw
#5 ("Neuromuscular"):ti,ab,kw
#6 ("Balance"):ti,ab,kw
#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6
#8 ("Training"):ti,ab,kw
#9 #7 AND #8
#10 [Meniscectomy] explode all trees
#11 ("Meniscectomies"):ti,ab,kw OR ("Arthroscopic Meniscectom*"):ti,ab,kw OR ("Meniscectom*, Arthroscopic"):ti,ab,kw OR ("Meniscal Resection*"):ti,ab,kw OR ("Resection*, Meniscal"):ti,ab,kw
#12 #10 OR #11
#13 #9 AND #12

Search strategy for Web of science

#1 TS=(Propriocept* OR Position Sense* OR Sense*, Position OR Sense of Position OR Sensorimotor OR Proprioceptive neuromuscular facilitation OR Neuromuscular OR Balance)
#2 TS= (Training)
#3 TS=(Meniscectom* OR Arthroscopic Meniscectom* OR Meniscectom*, Arthroscopic OR Meniscal Resection* OR Resection*, Meniscal)
#4 (#1 AND #2 AND #3)

Search strategy for China National Knowledge Infrastructure (CNKI)

#1 "本体感觉"[主题]
#2 "平衡"[主题]
#3 "神经肌肉"[主题]
#4 "本体感觉神经肌肉促进技术"[主题]
#5 "感觉运动"[主题]
#6 "训练"[主题]
#7 "半月板术后"[主题]
#8 "半月板切除术"[主题]
#9 "半月板成型术"[主题]
#10 "半月板修整术"[主题]
#11 #1 OR #2 OR #3 OR #4 OR #1
#12 #7 OR #8 OR #9 OR #10

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#13 #6 AND #11 AND #12

Search strategy for Technology Periodical Database (VIP)

#1 “本体感觉”[题名或关键词]
#2 “平衡”[题名或关键词]
#3 “神经肌肉”[题名或关键词]
#4 “本体感觉神经肌肉促进技术”[题名或关键词]
#5 “感觉运动”[题名或关键词]
#6 “训练”[题名或关键词]
#7 “半月板术后”[题名或关键词]
#8 “半月板切除术”[题名或关键词]
#9 “半月板成型术”[题名或关键词]
#10 “半月板修整术”[题名或关键词]
#11 #1 OR #2 OR #3 OR #4 OR #1
#12 #7 OR #8 OR #9 OR #10
#13 #6 AND #11 AND #12

Search strategy for WanFang Data

#1 “本体感觉”[主题]
#2 “平衡”[主题]
#3 “神经肌肉”[主题]
#4 “本体感觉神经肌肉促进技术”[主题]
#5 “感觉运动”[主题]
#6 “训练”[主题]
#7 “半月板术后”[主题]
#8 “半月板切除术”[主题]
#9 “半月板成型术”[主题]
#10 “半月板修整术”[主题]
#11 #1 OR #2 OR #3 OR #4 OR #1
#12 #7 OR #8 OR #9 OR #10
#13 #6 AND #11 AND #12

Search strategy for China Biology Medicine (CBM)

#1 “本体感觉”[常用字段:智能]
#2 “平衡”[常用字段:智能]
#3 “神经肌肉”[常用字段:智能]
#4 “本体感觉神经肌肉促进技术”[常用字段:智能]
#5 “感觉运动”[常用字段:智能]
#6 “训练”[常用字段:智能]
#7 “半月板术后”[常用字段:智能]
#8 “半月板切除术”[常用字段:智能]
#9 “半月板成型术”[常用字段:智能]
#10 “半月板修整术”[常用字段:智能]
#11 #1 OR #2 OR #3 OR #4 OR #1

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#12 #7 OR #8 OR #9 OR #10
#13 #6 AND #11 AND #12

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	3-4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	5-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	6,7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	7
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	7
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	6,7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	7
Study characteristics	17	Cite each included study and present its characteristics.	8
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	8
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	8
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	8-9
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	8-9
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	8-9
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	8-9
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	-
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	9-10
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	10-12
	23b	Discuss any limitations of the evidence included in the review.	12-13
	23c	Discuss any limitations of the review processes used.	12-13
	23d	Discuss implications of the results for practice, policy, and future research.	13
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	4-5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	4-5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	4-5
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	13
Competing interests	26	Declare any competing interests of review authors.	14
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	14

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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